



**CLC**

Educational Institute

**DIGITAL SAT — MATH**

# Theory Companion — Part 1

## Algebra & Data Analysis

Lines | Systems | Inequalities | Absolute Value | Exponents | Radicals

Ratios & Proportions | Probability | Unit Conversions | Statistics | Data Displays | Scatterplots | Sampling

A shortened, high-yield version of CLC's full Math theory reference —

condensed rules for all 13 skills, each followed by 8 Hard/Very Hard worked problems broken into Formula, Solution, Tip, Trap, Process of Elimination, Shortcut, and Desmos Tip.

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## Digital SAT Math — Overview

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Math accounts for half of the Digital SAT score, split across Algebra, Advanced Math, Problem-Solving & Data Analysis, and Geometry & Trigonometry. This companion (Part 1) covers 13 skills from Algebra and Problem-Solving & Data Analysis — the ones built on proportional reasoning, linear relationships, and statistical thinking. Each skill is condensed to its essential rules, then drilled with 8 genuinely difficult problems built around the specific twists the real exam uses to separate students who memorized a formula from students who understand it.

| Group                           | Skills Covered                                                                                                                                                 |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Algebra                         | Lines · Systems of Linear Equations · Linear Inequalities · Absolute Value · Exponent Rules · Radicals, Rational Exponents & Scientific Notation               |
| Problem-Solving & Data Analysis | Ratios & Proportions · Probability · Unit Conversions & Rates · Statistics — Center & Spread · Data Displays · Scatterplots · Sampling, Bias & Margin of Error |

This is deliberately a SHORTENED companion: the full theory reference (with every derivation, extra example, and edge case) exists separately. The purpose here is speed and critical thinking under time pressure — recognizing which rule applies when a question is disguised, combined with another rule, or padded with irrelevant information, the way the actual Digital SAT writes Hard and Very Hard questions.

Every worked problem follows the same seven-part breakdown used across CLC's companion series: the Formula/Rule the problem hinges on, a full step-by-step Solution, a Tip (the reliable first move for this problem type), a Trap (the specific wrong turn this problem is built to tempt), a Process of Elimination table naming the concrete wrong answers a rushed student lands on and exactly which slip produces each one, a Shortcut where one genuinely exists, and a Desmos Tip showing how to confirm or accelerate the solution on the Digital SAT's built-in graphing calculator.

## PART 1 — ALGEBRA

### 1. Lines

#### Lines — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

| Term (English)       | Greek — Ελληνικά            |
|----------------------|-----------------------------|
| Slope                | Κλίση                       |
| Y-intercept          | Σημείο τομής με τον άξονα y |
| Slope-intercept form | Μορφή κλίσης-τομής          |
| Parallel lines       | Παράλληλες ευθείες          |
| Perpendicular lines  | Κάθετες ευθείες             |
| Point-slope form     | Μορφή σημείου-κλίσης        |

#### Lines — Condensed Theory

Linear functions model any relationship with a constant rate of change. Every Lines question comes down to slope, one of four equation forms, or a small set of coordinate-geometry formulas.

##### Slope Formula

$$m = (y_2 - y_1) / (x_2 - x_1)$$

Positive slope rises left to right; negative slope falls. A horizontal line has slope 0; a vertical line has undefined slope.

##### Slope-Intercept Form

$$y = mx + b$$

m is the slope (rate of change); b is the y-intercept (value of y when x = 0).

##### Point-Slope Form

$$y - y_1 = m(x - x_1)$$

Use when you know the slope and one point; solve for y to convert to slope-intercept form.

##### Standard Form

$$Ax + By = C \rightarrow \text{slope} = -A/B, \text{ y-intercept} = C/B$$



Always convert to slope-intercept form to read off the slope or y-intercept directly.

### Parallel & Perpendicular Lines

**Parallel:**  $m_1 = m_2$ .    **Perpendicular:**  $m_1 \times m_2 = -1$

Perpendicular slopes are negative reciprocals of each other.

### Intercepts

**x-intercept:** set  $y = 0$ .    **y-intercept:** set  $x = 0$ .

Solve the resulting one-variable equation.

### Midpoint Formula

$$M = ((x_1+x_2)/2, (y_1+y_2)/2)$$

Average the x-coordinates and the y-coordinates separately.

### Distance Formula

$$d = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$$

Derived from the Pythagorean theorem; can also be found by drawing the right triangle between the two points.



### Watch Out

Never count grid squares to estimate slope from a graph — the axes are often scaled differently. Always use the actual coordinate values.

The SAT deliberately writes many Lines questions so that graphing alone won't get you the exact answer — know the algebra, don't just rely on Desmos.

## Lines — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

### Lines — Problem 1 (Very Hard)



Segment AB has endpoints A(-2, 5) and B(6, 1). Line m is the perpendicular bisector of segment AB. What is the y-intercept of line m?

**Formula / Rule Used:** Midpoint formula and negative reciprocal slopes (perpendicular bisector).

**Solution**

**Step 1:** Find the midpoint of segment AB:  $M = ((-2+6)/2, (5+1)/2) = (4/2, 6/2) = (2, 3)$ .

**Step 2:** Find the slope of AB:  $m_{AB} = (1 - 5)/(6 - (-2)) = -4/8 = -1/2$ .

**Step 3:** Line m is the perpendicular bisector, so its slope is the negative reciprocal of  $-1/2$ , which is 2.

**Step 4:** Line m passes through the midpoint (2, 3) with slope 2. Use point-slope form:  $y - 3 = 2(x - 2)$ .

**Step 5:** Simplify:  $y - 3 = 2x - 4$ , so  $y = 2x - 1$ .

**Step 6:** The y-intercept occurs where  $x = 0$ :  $y = 2(0) - 1 = -1$ .

**Answer:** y-intercept = -1, so line m crosses the y-axis at (0, -1).

**Tip**

For a perpendicular bisector problem, always compute the midpoint AND the perpendicular slope before writing the equation -- both pieces are required.

**Trap**

A common error is building the bisector line through one of the original endpoints (A or B) instead of the midpoint, which produces a different, incorrect y-intercept.

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| y-intercept = 9       | Used endpoint A(-2, 5) instead of the midpoint (2, 3) as the point for the bisector line, keeping the correct perpendicular slope of 2 but building the equation through the wrong point. |
| y-intercept = 4       | Used the slope of segment AB itself ( $-1/2$ ) instead of its negative reciprocal (2) when writing the bisector's equation through the midpoint.                                          |
| y-intercept = 7       | Made a sign error finding the slope of AB, computing $(1-5)/(6+2)$ as $4/8 = 1/2$ instead of $-4/8 = -1/2$ , which flipped the perpendicular slope to -2 instead of 2.                    |

**Shortcut**

Since only the y-intercept is needed, you can skip fully simplifying to standard form -- just solve the point-slope equation for y and read off the constant term directly.

**Desmos Tip**

Plot A(-2,5) and B(6,1), graph the segment, then graph  $y = 2x - 1$  and confirm it crosses AB at a right angle exactly at the midpoint (2,3).



Line p satisfies the equation  $3(2y - x) = 4(y + 6)$ . Line q is parallel to line p and passes through the point (4, -1). Write the equation of line q in standard form  $Ax + By = C$ , where A, B, and C are integers and A is positive.

**Formula / Rule Used:** Simplify to slope-intercept form; parallel lines share slope; convert to standard form.

**Solution**

**Step 1:** Distribute both sides of line p's equation:  $6y - 3x = 4y + 24$ .

**Step 2:** Collect y-terms on one side:  $6y - 4y = 3x + 24$ , so  $2y = 3x + 24$ .

**Step 3:** Solve for y:  $y = (3/2)x + 12$ , so line p has slope  $3/2$ .

**Step 4:** Line q is parallel to line p, so line q also has slope  $3/2$ .

**Step 5:** Use point-slope form with (4, -1):  $y - (-1) = (3/2)(x - 4)$ , so  $y + 1 = (3/2)(x - 4)$ .

**Step 6:** Multiply both sides by 2 to clear the fraction:  $2y + 2 = 3(x - 4) = 3x - 12$ .

**Step 7:** Rearrange into standard form:  $3x - 2y = 14$ .

**Answer:**  $3x - 2y = 14$

**Tip**

When an equation is wrapped in parentheses or mixes x and y on both sides, distribute and collect like terms first -- don't try to read off the slope before it's isolated.

**Trap**

A rushed student distributes only part of the equation or mishandles the coefficient when isolating y, landing on slope 3 instead of the correct slope  $3/2$ , which produces the wrong final equation for line q.

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $3x - y = 13$         | Mishandled isolating y from $2y = 3x + 24$ , dividing only the x-term by 2 and not the constant, landing on slope 3 instead of $3/2$ , then building line q's equation with that wrong slope. |
| $3x - 2y = 10$        | Used the correct slope $3/2$ but dropped the negative sign on -1 when substituting the point, writing $y - 1 = (3/2)(x - 4)$ instead of $y - (-1) = (3/2)(x - 4)$ .                           |
| $3x - 2y = 6$         | Forgot to multiply the constant by 3 when distributing on the right side, writing $3(x - 4)$ as $3x - 4$ instead of $3x - 12$ , which shifted the final constant term.                        |

**Shortcut**

Once you know line q needs the same slope as line p, you can skip fully solving line p for y -- just read the slope directly as the ratio of coefficients from  $2y = 3x + 24$  ( $3/2$ ) without finishing the simplification.

**Desmos Tip**

Graph  $3(2y - x) = 4(y + 6)$  directly (Desmos accepts implicit equations) alongside  $3x - 2y = 14$  and confirm the two lines never intersect and that the second passes through (4, -1).

**Lines — Problem 3** (Very Hard)

Point P has coordinates  $(k, 3)$ , and point Q has coordinates  $(-2, -3)$ . The distance between P and Q is 10 units. What is the sum of all possible values of  $k$ ?

**Formula / Rule Used:** Distance formula, solved as a quadratic for an unknown coordinate.

**Solution**

**Step 1:** Apply the distance formula:  $d = \sqrt{(k - (-2))^2 + (3 - (-3))^2}$ .

**Step 2:** Substitute  $d = 10$ :  $10 = \sqrt{(k + 2)^2 + 6^2}$ .

**Step 3:** Square both sides:  $100 = (k + 2)^2 + 36$ .

**Step 4:** Isolate the squared term:  $(k + 2)^2 = 64$ .

**Step 5:** Take the square root of both sides, keeping both signs:  $k + 2 = 8$  or  $k + 2 = -8$ .

**Step 6:** Solve each case:  $k = 6$  or  $k = -10$ .

**Step 7:** Add the two possible values:  $6 + (-10) = -4$ .

**Answer:** -4

**Tip**

Whenever a distance-formula problem gives you the distance and asks for a coordinate, expect a squared equation with two valid solutions -- solve it like any other quadratic and keep both roots unless the context rules one out.

**Trap**

Many students take the square root of  $(k+2)^2 = 64$  and keep only the positive root ( $k = 6$ ), missing  $k = -10$  entirely and reporting an incomplete answer.

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6                     | Took the square root of $(k+2)^2 = 64$ and kept only the positive root ( $k+2 = 8$ , so $k = 6$ ), discarding the negative branch entirely and reporting that single value instead of the sum of both roots. |
| 4                     | Misread the x-coordinate of Q as +2 instead of -2, so assumed the two $k$ -values were symmetric about $k = 2$ and reported their sum as $2(2) = 4$ instead of $2(-2) = -4$ .                                |
| -60                   | Correctly found both roots $k = 6$ and $k = -10$ , but multiplied them together instead of adding them, computing the product $(-60)$ instead of the requested sum $(-4)$ .                                  |

**Shortcut**

Because the problem only asks for the SUM of the possible  $k$ -values, you don't need to solve both equations individually: the two roots of  $(k+2)^2 = 64$  are symmetric around  $k = -2$ , so their sum is always  $2(-2) = -4$ .

**Desmos Tip**



Graph the circle  $(x + 2)^2 + (y + 3)^2 = 100$  centered at Q with radius 10, then find where it crosses the horizontal line  $y = 3$  -- the x-coordinates of the two intersection points are the two values of k.

### Lines — Problem 4 (Hard)

A bike-rental kiosk in Rotterdam charges a fixed unlocking fee plus a constant rate per minute. Nikos rents a bike for 15 minutes and pays 4.75 euros total. His friend Elena rents a bike for 40 minutes and pays 9.00 euros total. Assuming cost is a linear function of rental time, how much would a 70-minute rental cost, in euros?

**Formula / Rule Used:** Build a linear model  $C = mt + b$  from two data points, then evaluate it.

#### Solution

**Step 1:** Translate the two rentals into coordinate points (time, cost): (15, 4.75) and (40, 9.00).

**Step 2:** Find the rate (slope) using these points:  $m = (9.00 - 4.75)/(40 - 15) = 4.25/25 = 0.17$  euros per minute.

**Step 3:** Use point-slope form with (15, 4.75) to find the fixed fee (y-intercept):  $4.75 = 0.17(15) + b$ .

**Step 4:** Compute  $0.17(15) = 2.55$ , so  $4.75 = 2.55 + b$ , giving  $b = 2.20$  euros.

**Step 5:** Write the full cost model:  $C = 0.17t + 2.20$ .

**Step 6:** Substitute  $t = 70$ :  $C = 0.17(70) + 2.20 = 11.90 + 2.20$ .

**Step 7:** Add:  $C = 14.10$  euros.

**Answer:** 14.10 euros

#### Tip

In a linear word problem, immediately convert each described scenario into an (input, output) ordered pair -- once you have two points, it's a standard slope-and-intercept problem.

#### Trap

A student who only computes the per-minute rate and multiplies it by 70 (getting  $0.17 \times 70 = 11.90$ ) forgets the fixed unlocking fee entirely and arrives at the wrong total.

#### Process of Elimination

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                    |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11.90 euros           | Computed only the per-minute rate (0.17) and multiplied it by 70, forgetting to add the fixed unlocking fee of 2.20 euros.                                                                 |
| 15.75 euros           | Rounded the per-minute rate from 0.17 to 0.2 too early, then carried that rounding error through the intercept ( $b = 1.75$ ) and the final calculation instead of keeping the exact rate. |
| 11.55 euros           | Misread the question as asking for the cost of 70 additional minutes beyond Nikos's original 15-minute rental, substituting $t = 55$ instead of $t = 70$ into the cost model.              |

#### Shortcut



No shortcut beats setting up the two points and solving for  $m$  and  $b$  directly -- this is already the fastest reliable path for a two-point linear word problem.

### Desmos Tip

Enter the two points (15, 4.75) and (40, 9.00) in a Desmos table, graph  $y = 0.17x + 2.20$  through them, and read the  $y$ -value at  $x = 70$  directly off the graph.

## Lines — Problem 5 (Hard)

Triangle ABC has vertices A(-4, -3), B(6, -3), and C(2, 9). Side AB is horizontal, and the slope of side BC is -3. Find the perimeter of triangle ABC, in exact simplified radical form.

**Formula / Rule Used:** Distance formula applied to each side of a triangle to find perimeter.

### Solution

**Step 1:** Since A and B share the same  $y$ -coordinate (-3), side AB is horizontal, so its length is simply the difference in  $x$ -coordinates:  $AB = |6 - (-4)| = 10$ .

**Step 2:** The slope of BC is given but not needed to find lengths -- apply the distance formula directly to B(6, -3) and C(2, 9):  $BC = \sqrt{(2-6)^2 + (9-(-3))^2}$ .

**Step 3:** Compute inside the square root:  $(2-6)^2 = 16$  and  $(9-(-3))^2 = 144$ , so  $BC = \sqrt{16 + 144} = \sqrt{160}$ .

**Step 4:** Simplify  $\sqrt{160} = \sqrt{16 \times 10} = 4\sqrt{10}$ .

**Step 5:** Apply the distance formula to A(-4, -3) and C(2, 9):  $AC = \sqrt{(2-(-4))^2 + (9-(-3))^2}$ .

**Step 6:** Compute inside the square root:  $(2-(-4))^2 = 36$  and  $(9-(-3))^2 = 144$ , so  $AC = \sqrt{36 + 144} = \sqrt{180}$ .

**Step 7:** Simplify  $\sqrt{180} = \sqrt{36 \times 5} = 6\sqrt{5}$ .

**Step 8:** Add all three sides for the perimeter:  $\text{Perimeter} = 10 + 4\sqrt{10} + 6\sqrt{5}$ .

**Answer:**  $10 + 4\sqrt{10} + 6\sqrt{5}$  units (approximately 36.0 units)

### Tip

For a perimeter or side-length problem, apply the distance formula to each pair of vertices directly -- don't try to use slope information, which only tells you direction, not length.

### Trap

The slope of BC (-3) is a tempting piece of data, and a student might try to fold it into a length calculation (treating it as a ratio of legs), which doesn't give the actual side length and wastes time or produces a wrong value.

### Process of Elimination

| Tempting Wrong Answer                                          | Exactly What Went Wrong                                                                                                                                     |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $2 + 4\sqrt{10} + 6\sqrt{5}$ units (approximately 28.1 units)  | Misread vertex A's $x$ -coordinate as 4 instead of -4, computing $AB =  6 - 4  = 2$ instead of the correct $AB =  6 - (-4)  = 10$ .                         |
| $10 + 4\sqrt{10} + 3\sqrt{2}$ units (approximately 27.9 units) | When computing AC, added the coordinate differences directly instead of squaring them first: $\sqrt{6 + 12} = \sqrt{18} = 3\sqrt{2}$ , instead of correctly |



|                                                           |                                                                                                                              |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
|                                                           | squaring to get $\sqrt{36 + 144} = \sqrt{180} = 6\sqrt{5}$ .                                                                 |
| <b>10 + 4sqrt(10) units</b><br>(approximately 22.6 units) | Computed AB and BC correctly but forgot to include side AC in the perimeter sum, stopping after only two of the three sides. |

**Shortcut**

Recognizing that AB is horizontal (equal y-coordinates) lets you find that side by simple subtraction of x-values instead of running the full distance formula, saving a step on one of the three sides.

**Desmos Tip**

Plot the three vertices in Desmos, draw the triangle with the polygon feature, and use the distance measurement between each pair of points to confirm  $BC = \sqrt{160}$  and  $AC = \sqrt{180}$  numerically.

**Lines — Problem 6** (Very Hard)

Line r is given by  $kx + 3y = 12$ , and line s is given by  $2x - 5y = 10$ . If line r is perpendicular to line s, what is the x-intercept of line r?

**Formula / Rule Used:** Standard-form slope ( $-A/B$ ), perpendicular negative reciprocal, then x-intercept.

**Solution**

**Step 1:** Find the slope of line s from standard form  $Ax + By = C$ , where slope =  $-A/B$ : slope of s =  $-2/(-5) = 2/5$ .

**Step 2:** Since line r is perpendicular to line s, its slope must be the negative reciprocal of  $2/5$ , which is  $-5/2$ .

**Step 3:** Find the slope of line r in terms of k from its standard form  $kx + 3y = 12$ : slope of r =  $-k/3$ .

**Step 4:** Set the two expressions for line r's slope equal:  $-k/3 = -5/2$ .

**Step 5:** Solve for k:  $k/3 = 5/2$ , so  $k = 15/2$ .

**Step 6:** Substitute  $k = 15/2$  back into line r's equation:  $(15/2)x + 3y = 12$ .

**Step 7:** Multiply every term by 2 to clear the fraction:  $15x + 6y = 24$ , which simplifies (divide by 3) to  $5x + 2y = 8$ .

**Step 8:** Find the x-intercept by setting  $y = 0$ :  $5x = 8$ , so  $x = 8/5$ .

**Answer:** x-intercept =  $(8/5, 0)$

**Tip**

When a line's equation contains an unknown coefficient and a perpendicularity or parallelism condition, first write both slopes in terms of A and B (slope =  $-A/B$  for standard form), then set up one equation to solve for the unknown before touching the rest of the problem.

**Trap**

A student who confuses the perpendicular condition with the parallel condition sets  $-k/3 = 2/5$  (matching slopes instead of using negative reciprocals), which yields the wrong value of k and a wrong final intercept.

**Process of Elimination**



| Tempting Wrong Answer          | Exactly What Went Wrong                                                                                                                                                                                         |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>x-intercept = (-10, 0)</b>  | Confused the perpendicular condition with the parallel condition, setting $-k/3 = 2/5$ (matching slopes) instead of $-k/3 = -5/2$ (negative reciprocal), which gave $k = -6/5$ and a completely different line. |
| <b>x-intercept = (72/5, 0)</b> | Correctly set up $k/3 = 5/2$ but solved it by dividing $5/2$ by 3 instead of multiplying by 3, getting $k = 5/6$ instead of the correct $k = 15/2$ .                                                            |
| <b>x-intercept = (0, 4)</b>    | Found the correct simplified line $5x + 2y = 8$ but set $x = 0$ instead of $y = 0$ , computing the y-intercept (0, 4) and mislabeling it as the x-intercept.                                                    |

### Shortcut

There isn't a faster route than solving for  $k$  first -- the x-intercept genuinely depends on  $k$ , so both steps (solve for  $k$ , then find the intercept) are necessary in this order.

### Desmos Tip

After solving for  $k$ , graph  $5x + 2y = 8$  and  $2x - 5y = 10$  together in Desmos and confirm visually that the lines cross at a right angle and that the first line crosses the x-axis at  $8/5 = 1.6$ .

## Lines — Problem 7 (Very Hard)

**For what value of  $k$  does the system of equations  $4x - 6y = 10$  and  $kx + 9y = -20$  have no solution?**

**Formula / Rule Used:** Standard-form slope ( $-A/B$ ); parallel lines share slope but must have different constants for a true 'no solution' case.

### Solution

**Step 1:** Find the slope of the first line from standard form:  $4x - 6y = 10$  has slope  $= -A/B = -4/(-6) = 2/3$ .

**Step 2:** For the system to have no solution, the two lines must be parallel (equal slopes) but not identical. Find the slope of the second line in terms of  $k$ :  $kx + 9y = -20$  has slope  $= -k/9$ .

**Step 3:** Set the slopes equal:  $-k/9 = 2/3$ .

**Step 4:** Solve for  $k$ : multiply both sides by 9:  $-k = 6$ , so  $k = -6$ .

**Step 5:** Check that the lines aren't identical when  $k = -6$  (which would give infinitely many solutions instead of no solution): substitute  $k = -6$ , so the second line becomes  $-6x + 9y = -20$ .

**Step 6:** Scale the first line ( $4x - 6y = 10$ ) by  $-1.5$  to match the second line's x-coefficient:  $-6x + 9y = -15$ .

**Step 7:** Since  $-15$  does not equal  $-20$ , the two lines are parallel but distinct, confirming the system truly has no solution rather than infinitely many.

**Step 8:** Therefore  $k = -6$ .

**Answer:  $k = -6$**

### Tip

Whenever a system-of-equations problem asks for 'no solution', translate it as 'same slope, different intercept' -- find both lines' slopes from standard form using  $-A/B$  and set them equal.

### Trap



The most common error is misreading the standard-form slope formula  $-A/B$  when  $B$  is already negative (here  $B = -6$ ): students often compute  $-4/6 = -2/3$  instead of correctly canceling the double negative to get  $2/3$ , which flips the sign of the final answer.

### Process of Elimination

| Tempting Wrong Answer        | Exactly What Went Wrong                                                                                                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><math>k = 6</math></b>    | Miscomputed the slope of the first line from standard form as $-4/6 = -2/3$ (mishandling the double negative from $B = -6$ ), instead of the correct $2/3$ , which flipped the sign of the final answer. |
| <b><math>k = 13.5</math></b> | Used the perpendicular condition (product of slopes $= -1$ ) instead of the parallel/no-solution condition (equal slopes), solving $(2/3)(-k/9) = -1$ instead of $-k/9 = 2/3$ .                          |
| <b><math>k = -9</math></b>   | Made an arithmetic slip computing $9 \times (2/3)$ , getting 9 instead of 6, so $-k = 9$ was solved as $k = -9$ instead of the correct $k = -6$ .                                                        |

### Shortcut

Skip computing each slope separately -- cross-multiply the coefficients directly (parallel condition  $A_1 \times B_2 = A_2 \times B_1$ ):  $4(9) = k(-6)$ , so  $36 = -6k$ , giving  $k = -6$  in one line.

### Desmos Tip

Graph  $4x - 6y = 10$  and  $kx + 9y = -20$  for a few trial  $k$ -values near  $-6$  and watch the second line rotate until it looks parallel to the first, then confirm  $k = -6$  makes the lines never touch.

## Lines — Problem 8 (Hard)

Line  $w$  passes through the points  $(2, 5)$  and  $(6, -3)$ . Line  $w$  crosses the  $x$ -axis at point  $R$  and the  $y$ -axis at point  $S$ . What is the area of triangle  $ORS$ , where  $O$  is the origin?

**Formula / Rule Used:** Slope formula, point-slope form,  $x$ - and  $y$ -intercepts, triangle area  $= (1/2)(\text{base})(\text{height})$ .

### Solution

**Step 1:** Find the slope of line  $w$  using the two given points, keeping the same point order in numerator and denominator:  $m = (-3 - 5)/(6 - 2) = -8/4 = -2$ .

**Step 2:** Use point-slope form with  $(2, 5)$ :  $y - 5 = -2(x - 2)$ .

**Step 3:** Simplify:  $y - 5 = -2x + 4$ , so  $y = -2x + 9$ .

**Step 4:** Find the  $y$ -intercept (point  $S$ ) by setting  $x = 0$ :  $y = -2(0) + 9 = 9$ , so  $S = (0, 9)$ .

**Step 5:** Find the  $x$ -intercept (point  $R$ ) by setting  $y = 0$ :  $0 = -2x + 9$ , so  $2x = 9$ ,  $x = 9/2 = 4.5$ , so  $R = (4.5, 0)$ .

**Step 6:** Triangle  $ORS$  is a right triangle with legs  $OS$  (along the  $y$ -axis, length 9) and  $OR$  (along the  $x$ -axis, length 4.5), since the axes meet at the origin at a right angle.

**Step 7:** Area  $= (1/2)(\text{base})(\text{height}) = (1/2)(4.5)(9) = (1/2)(40.5) = 20.25$ .

**Answer:** Area = 20.25 square units ( $81/4$ ).

**Tip**

When a triangle is formed by a line and both coordinate axes, find the line's x- and y-intercepts first -- those two intercepts are automatically the legs of a right triangle with the origin, so the area formula applies directly.

**Trap**

The most common slip is losing track of point order when computing slope from two points -- pairing the y-values in one order but the x-values in the reverse order silently flips the sign of the slope, which then shifts both intercepts and produces a completely different (and much smaller) triangle.

**Process of Elimination**

| Tempting Wrong Answer                 | Exactly What Went Wrong                                                                                                                                                                                                         |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Area = 0.25 (1/4) square units</b> | Mismatched the point order in the slope formula, computing $m = (-3-5)/(2-6) = 2$ instead of pairing coordinates in the same order ( $m = (-3-5)/(6-2) = -2$ ), which flipped the sign of the slope and shrank both intercepts. |
| <b>Area = 40.5 square units</b>       | Found both intercepts correctly ( $R = (4.5, 0)$ , $S = (0, 9)$ ) but forgot to multiply by $1/2$ in the triangle area formula, computing base $\times$ height instead of $(1/2)(\text{base})(\text{height})$ .                 |
| <b>Area = 22.5 square units</b>       | Correctly found the line $y = -2x + 9$ , but rounded the x-intercept $9/2 = 4.5$ up to 5 instead of keeping the exact value, then computed $(1/2)(5)(9) = 22.5$ instead of $(1/2)(4.5)(9) = 20.25$ .                            |

**Shortcut**

You can't skip any of the three formulas here, but you can save a step by reading the y-intercept  $b$  directly from  $y = mx + b$  using either point ( $5 = -2(2) + b$ , so  $b = 9$ ) instead of writing out full point-slope form first.

**Desmos Tip**

Graph the line through (2,5) and (6,-3) in Desmos -- it will automatically display where the line crosses each axis, so you can read  $R$  and  $S$  directly and check the area with Desmos's polygon/area tool on triangle  $O, R, S$ .

## 2. Systems of Linear Equations

### Systems of Linear Equations — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

| Term (English)                   | Greek — Ελληνικά            |
|----------------------------------|-----------------------------|
| System of equations              | Σύστημα εξισώσεων           |
| Solution (of a system)           | Λύση συστήματος             |
| Substitution method              | Μέθοδος αντικατάστασης      |
| Elimination method               | Μέθοδος απαλοιφής           |
| No solution / infinite solutions | Καμία λύση / άπειρες λύσεις |

### Systems of Linear Equations — Condensed Theory

A system of two linear equations is solved where their graphs intersect. The SAT tests both finding that solution and reasoning about how many solutions exist without fully solving.

#### Substitution

Isolate one variable in one equation, substitute it into the other, solve, then back-substitute. Best when a variable already has coefficient 1.

#### Elimination

Multiply one or both equations so a variable's coefficients become opposites, then add the equations to cancel that variable.

#### Setting Equal

$$f(x) = g(x)$$

If both equations are already solved for y, set the right-hand sides equal and solve for x directly.

#### Solving Directly for an Expression (shortcut)

If asked for a combined value like  $2x + 3y$  rather than x and y individually, check whether adding/subtracting/scaling the two equations produces that expression directly — this is almost always faster than solving the full system.

#### Number of Solutions

**One:**  $m_1 \neq m_2$ . **None:**  $m_1 = m_2$ ,  $b_1 \neq b_2$ . **Infinite:**  $m_1 = m_2$ ,  $b_1 = b_2$ .

In standard form  $Ax + By = C$ , no solution means  $A_1/A_2 = B_1/B_2 \neq C_1/C_2$ ; infinite solutions means  $A_1/A_2 =$



$$B1/B2 = C1/C2.$$

### ⚠ Watch Out

For a 'no solution' or 'infinite solutions' question with an unknown constant, use the coefficient-ratio shortcut instead of solving the whole system — it turns the problem into a simple proportion.

## Systems of Linear Equations — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

### Systems of Linear Equations — Problem 1 (Very Hard)

Consider the system  $2x = 11 - 5y$  and  $ky = 30 - 6x$ , where  $k$  is a constant. For what value of  $k$  does this system have no solution?

**Formula / Rule Used:** No-solution condition for a linear system:  $A1/A2 = B1/B2$  does not equal  $C1/C2$  in standard form  $Ax + By = C$ .

#### Solution

**Step 1:** Rewrite the first equation in standard form:  $2x + 5y = 11$ .

**Step 2:** Rewrite the second equation in standard form:  $6x + ky = 30$ .

**Step 3:** For no solution, the coefficient ratios must match:  $2/6 = 5/k$ .

**Step 4:** Cross-multiply:  $2k = 30$ , so  $k = 15$ .

**Step 5:** Check that the constant ratio is genuinely different (required for no solution, not infinite solutions):  $C1/C2 = 11/30$ , while  $A1/A2 = 2/6 = 1/3 = 10/30$ .

**Step 6:** Since  $11/30$  does not equal  $10/30$ , the lines are parallel but distinct at  $k = 15$ , confirming no solution.

**Answer:**  $k = 15$

### Tip

Before comparing ratios, rearrange every equation into  $Ax + By = C$  form -- don't try to compare coefficients while variables and constants are mixed on both sides.

### Trap

A student who finds  $k = 15$  from the coefficient ratio alone often stops and assumes matching slopes automatically means infinitely many solutions; they must also check that the constant ratio ( $11/30$ ) differs from the coefficient ratio ( $1/3$ ) to confirm this is actually a no-solution case.

### Process of Elimination

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $k = -15$             | Rearranged $ky = 30 - 6x$ incorrectly as $6x - ky = 30$ (sign error on the $y$ -term) instead of $6x + ky = 30$ , which flips the sign in the ratio and produces |



|            |                                                                                                                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | $k = -15$ instead of $k = 15$ .                                                                                                                                                                                                     |
| $k = 12/5$ | Mispaired the coefficients when writing the ratio, using $5/6 = 2/k$ (y-coefficient of equation 1 over x-coefficient of equation 2) instead of the correct $2/6 = 5/k$ , mixing up which coefficients correspond to which equation. |
| $k = 5/3$  | Cross-multiplied the proportion $2/6 = 5/k$ backwards, computing $2 \times 5 = 6 \times k$ instead of $2 \times k = 6 \times 5$ , giving $k = 5/3$ instead of 15.                                                                   |

**Shortcut**

Once both equations are in standard form, skip solving the system entirely -- just cross-multiply the x- and y-coefficient ratio ( $2/6 = 5/k$ ) to get  $k = 15$  in one line.

**Desmos Tip**

Graph  $2x + 5y = 11$  and  $6x + 15y = 30$  (solved for  $y$ ) in Desmos and confirm the two lines are visibly parallel and never touch.

**Systems of Linear Equations — Problem 2 (Hard)**

In the system below, the two lines intersect at a point located in Quadrant I of the  $xy$ -plane.  $2x + y = 9$  and  $x + 3y = 16$ . What is the value of  $3x + 4y$ ?

**Formula / Rule Used:** Linear combination shortcut: add or scale the two equations directly to build the target expression without solving for  $x$  and  $y$ .

**Solution**

**Step 1:** Notice the target expression  $3x + 4y$  can be built by adding the two equations as given:  $(2x + y) + (x + 3y) = 3x + 4y$ .

**Step 2:** Add the left sides:  $2x + y + x + 3y = 3x + 4y$ .

**Step 3:** Add the right sides:  $9 + 16 = 25$ .

**Step 4:** Therefore  $3x + 4y = 25$  directly, with no need to solve for  $x$  and  $y$  individually.

**Step 5:** As a check, solve the system fully: from equation 1,  $y = 9 - 2x$ ; substitute into equation 2:  $x + 3(9 - 2x) = 16$ , so  $x + 27 - 6x = 16$ , giving  $-5x = -11$  and  $x = 11/5$ .

**Step 6:** Then  $y = 9 - 2(11/5) = 9 - 22/5 = 23/5$ , and both values are positive, matching the Quadrant I detail (which is not needed to answer the question).

**Step 7:** Confirm:  $3(11/5) + 4(23/5) = 33/5 + 92/5 = 125/5 = 25$ , matching the shortcut answer.

**Answer:**  $3x + 4y = 25$

**Tip**

Before solving a system fully, check whether the target expression is already a simple sum, difference, or scaled combination of the two given equations -- if it is, add or subtract directly instead of finding  $x$  and  $y$ .

**Trap**



A student who sees a two-variable system and a geometric detail like 'Quadrant I' often assumes they must fully solve for  $x$  and  $y$  (landing on messy fractions like  $x = 11/5$ ) before computing  $3x + 4y$ , wasting time and risking an arithmetic slip -- the quadrant location is never actually used.

### Process of Elimination

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                                                           |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -7                    | Subtracted the two equations instead of adding them, computing $(2x + y) - (x + 3y) = 9 - 16 = -7$ , then mistakenly reported -7 as the value of $3x + 4y$ instead of recognizing only addition produces that target expression.  |
| -19                   | While solving the system fully as a check, distributed $3(9 - 2x)$ incorrectly as $27 - 2x$ instead of $27 - 6x$ , producing wrong values $x = 11$ and $y = -13$ , and an incorrect final combination of $3(11) + 4(-13) = -19$ . |
| 26                    | Misled by the Quadrant I detail into assuming the coordinates must be 'nice' integers, rounded $x = 11/5$ to 2 and used $y = 9 - 2(2) = 5$ , then computed $3(2) + 4(5) = 26$ instead of using the exact fractional solution.     |

### Shortcut

Just add the two equations as given -- their sum already equals  $3x + 4y$ , so the answer appears in one line without ever isolating  $x$  or  $y$ .

### Desmos Tip

Graph both lines in Desmos, find the intersection point  $(2.2, 4.6)$ , then compute  $3(2.2) + 4(4.6) = 25$  directly from the coordinates as a check.

## Systems of Linear Equations — Problem 3 (Very Hard)

Solve the system for the ordered pair  $(x, y)$ :  $x/2 + 5y/6 = 9/2$  and  $4x/3 - y = 7/3$ .

**Formula / Rule Used:** Clear fractions by multiplying each equation by its LCD, then eliminate a variable by multiplying both resulting equations by different constants.

### Solution

**Step 1:** Clear the first equation by multiplying every term by 6 (the LCD of 2 and 6):  $3x + 5y = 27$ .

**Step 2:** Clear the second equation by multiplying every term by 3 (the LCD of 3 and 1):  $4x - 3y = 7$ .

**Step 3:** To eliminate  $y$ , find a common multiple of 5 and 3 (which is 15): multiply the first cleared equation by 3 to get  $9x + 15y = 81$ .

**Step 4:** Multiply the second cleared equation by 5 to get  $20x - 15y = 35$ .

**Step 5:** Add the two new equations:  $9x + 15y + 20x - 15y = 81 + 35$ , so  $29x = 116$ .

**Step 6:** Solve for  $x$ :  $x = 116/29 = 4$ .

**Step 7:** Substitute  $x = 4$  into  $4x - 3y = 7$ :  $16 - 3y = 7$ , so  $-3y = -9$  and  $y = 3$ .

**Answer:**  $(x, y) = (4, 3)$

### Tip



Clear every fraction by multiplying each equation by its own LCD before attempting elimination -- never try to eliminate variables while denominators are still in play.

### Trap

A student often tries to eliminate  $y$  by scaling only one cleared equation (since 5 and -3 share no simple single multiplier), instead of multiplying BOTH cleared equations (by 3 and by 5) to reach matching  $y$ -coefficients of 15, which leads to leftover fractions and arithmetic errors.

### Process of Elimination

| Tempting Wrong Answer     | Exactly What Went Wrong                                                                                                                                                                                                                      |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $(x, y) = (4, -23/3)$     | After correctly finding $x = 4$ and substituting into $4x - 3y = 7$ to get $16 - 3y = 7$ , subtracted in the wrong direction (computed $-3y = 7 + 16$ instead of $-3y = 7 - 16$ ), giving $y = -23/3$ instead of 3.                          |
| $(x, y) = (3, 4)$         | Correctly computed the two numeric values 4 and 3 through elimination but swapped which value is $x$ and which is $y$ when writing the final ordered pair.                                                                                   |
| $(x, y) = (62/23, 87/23)$ | Cleared $4x/3 - y = 7/3$ by multiplying only the $x$ -term and the constant by 3, forgetting to also multiply the $y$ -term by 3, producing the wrong equation $4x - y = 7$ instead of $4x - 3y = 7$ , which throws off the entire solution. |

### Shortcut

No shortcut beats clearing fractions first -- trying to eliminate a variable while fractions remain multiplies the chance of arithmetic slips, so multiplying by the LCD immediately is itself the fastest path.

### Desmos Tip

Graph  $x/2 + 5y/6 = 9/2$  and  $4x/3 - y = 7/3$  directly in Desmos and use the intersection point to confirm the solution (4, 3).

## Systems of Linear Equations — Problem 4 (Hard)

The system  $5x - 2y = 16$  and  $bx + 3y = -1$  has a solution in which  $y = -3$ . What is the value of  $b - x$ ?

**Formula / Rule Used:** Substitute the known value into the equation without an unknown constant first, then use the result to solve for the unknown constant in the other equation.

### Solution

**Step 1:** Substitute  $y = -3$  into the first equation, which has no unknown constant:  $5x - 2(-3) = 16$ .

**Step 2:** Simplify:  $5x + 6 = 16$ , so  $5x = 10$  and  $x = 2$ .

**Step 3:** Substitute  $x = 2$  and  $y = -3$  into the second equation:  $b(2) + 3(-3) = -1$ .

**Step 4:** Simplify:  $2b - 9 = -1$ , so  $2b = 8$  and  $b = 4$ .

**Step 5:** Compute the requested expression:  $b - x = 4 - 2 = 2$ .

**Answer:**  $b - x = 2$

**Tip**

When one equation contains an unknown constant and the other doesn't, always use the equation without the unknown constant first to pin down the known variable before touching the equation that has two unknowns.

**Trap**

A student who plugs  $y = -3$  into the second equation first gets stuck with two unknowns ( $b$  and  $x$ ) in one equation and often guesses or mishandles the algebra; working equation 1 first avoids this entirely.

**Process of Elimination**

| <b>Tempting Wrong Answer</b> | <b>Exactly What Went Wrong</b>                                                                                                                                                                                          |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>-7</b>                    | Correctly found $x = 2$ but substituted $y = -3$ into the second equation with a sign flip, writing $2b + 3(3) = -1$ instead of $2b + 3(-3) = -1$ , getting $b = -5$ instead of $b = 4$ , then computed $-5 - 2 = -7$ . |
| <b>-2</b>                    | Correctly found $b = 4$ and $x = 2$ but computed $x - b$ instead of $b - x$ , reversing the order the question asks for.                                                                                                |
| <b>6</b>                     | Correctly found $b = 4$ and $x = 2$ but added the two values instead of subtracting, computing $b + x = 6$ instead of $b - x = 2$ .                                                                                     |

**Shortcut**

There's no way around the two substitutions here, but always resolving the 'clean' equation (the one without the unknown constant) first keeps you from ever juggling two unknowns at once.

**Desmos Tip**

After finding  $x = 2$ , graph  $bx + 3y = -1$  as  $y = (-1 - bx)/3$  and adjust  $b$  until the line passes through  $(2, -3)$  to visually confirm  $b = 4$ .

**Systems of Linear Equations — Problem 5 (Hard)**

Cycladic Ferries sells standard and express tickets for its Piraeus-Naxos route. The express ticket costs 15 euros more than twice the price of a standard ticket. Last week, a travel agency bought 3 standard tickets and 2 express tickets for a total of 205 euros. What is the combined price, in euros, of one standard ticket and one express ticket?

**Formula / Rule Used:** Translate verbal relationships into algebraic equations, then substitute to reduce the system to one variable.

#### Solution

**Step 1:** Let  $s$  = the price of a standard ticket and  $e$  = the price of an express ticket.

**Step 2:** Translate 'the express ticket costs 15 euros more than twice the standard price' into  $e = 2s + 15$ .

**Step 3:** Translate the purchase into an equation:  $3s + 2e = 205$ .

**Step 4:** Substitute  $e = 2s + 15$  into the second equation:  $3s + 2(2s + 15) = 205$ .

**Step 5:** Expand:  $3s + 4s + 30 = 205$ , so  $7s + 30 = 205$ .

**Step 6:** Solve for  $s$ :  $7s = 175$ , so  $s = 25$ .

**Step 7:** Find  $e$ :  $e = 2(25) + 15 = 65$ .

**Step 8:** Compute the combined price of one of each:  $s + e = 25 + 65 = 90$  euros.

**Answer:** 90 euros (standard = 25 euros, express = 65 euros)

#### Tip

Translate the comparison sentence into an equation first ( $e = 2s + 15$ ), then substitute it directly into the purchase equation instead of building two equations and eliminating -- it turns the system into a single-variable equation immediately.

#### Trap

A student may translate 'costs 15 euros more than twice the standard price' backwards as  $s = 2e + 15$  instead of  $e = 2s + 15$ , swapping which ticket is expressed in terms of the other and producing incorrect prices for both tickets.

#### Process of Elimination

| Tempting Wrong Answer     | Exactly What Went Wrong                                                                                                                                                                                                                 |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 75 euros                  | Translated the comparison sentence backwards as $s = 2e + 15$ instead of $e = 2s + 15$ , swapping which ticket price is expressed in terms of the other, giving standard = 55 and express = 20 (combined 75) instead of the correct 90. |
| 675/7 euros (about 96.43) | Substituted $e = 2s + 15$ into $3s + 2e = 205$ but distributed incorrectly as $3s + 4s + 15 = 205$ (forgot to double the 15), instead of the correct $3s + 4s + 30 = 205$ , throwing off both prices.                                   |
| 65 euros                  | Correctly solved for both prices ( $s = 25$ , $e = 65$ ) but reported only the express ticket price instead of computing the combined total $s + e = 90$ .                                                                              |

#### Shortcut

Since one equation already isolates  $e$  in terms of  $s$ , substitution is faster than elimination here -- there's no need to multiply or align coefficients at all.

**Desmos Tip**

Graph  $e = 2s + 15$  and  $3s + 2e = 205$  (solved for  $e$ ) in Desmos; the intersection point (25, 65) gives both ticket prices directly, and their sum confirms 90.

**Systems of Linear Equations — Problem 6** (Very Hard)

The system  $6x - 9y = 3k$  and  $8x - 12y = 20$  has infinitely many solutions, where  $k$  is a constant. What is the value of  $k^2 - 5$ ?

**Formula / Rule Used:** Infinite-solutions ratio shortcut:  $A_1/A_2 = B_1/B_2 = C_1/C_2$  for the two equations to represent the same line.

**Solution**

**Step 1:** Compare the x-coefficients:  $A_1/A_2 = 6/8 = 3/4$ .

**Step 2:** Compare the y-coefficients:  $B_1/B_2 = -9/-12 = 3/4$ , which matches, so the two lines already share a slope.

**Step 3:** For infinitely many solutions, the constant ratio must also equal  $3/4$ :  $3k/20 = 3/4$ .

**Step 4:** Cross-multiply:  $3k$  times  $4 = 3$  times  $20$ , so  $12k = 60$ .

**Step 5:** Solve for  $k$ :  $k = 5$ .

**Step 6:** Substitute into the requested expression:  $k^2 - 5 = 5^2 - 5 = 25 - 5 = 20$ .

**Answer:** 20

**Tip**

Whenever a system is described as having infinitely many solutions, immediately set up  $A_1/A_2 = B_1/B_2 = C_1/C_2$  instead of trying to 'solve' the system -- there is no unique point to find, only a constant to pin down.

**Trap**

A student who overlooks the coefficient 3 in front of  $k$  and solves  $6/8 = k/20$  instead of  $6/8 = 3k/20$  gets  $k = 15$  (not 5), which leads to a final answer of 220 instead of the correct 20.

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 220                   | Set up the constant ratio as $6/8 = k/20$ instead of $6/8 = 3k/20$ , forgetting the coefficient 3 in front of $k$ , getting $k = 15$ instead of 5, then computed $15^2 - 5 = 220$ .                   |
| 0                     | Correctly found $k = 5$ but misread the requested expression $k^2 - 5$ as $(k - 5)^2$ , computing $(5 - 5)^2 = 0$ instead of $5^2 - 5 = 20$ .                                                         |
| 5995/81 (about 74.0)  | Cross-multiplied the proportion $3k/20 = 3/4$ backward, computing $3k \times 3 = 20 \times 4$ instead of $3k \times 4 = 20 \times 3$ , giving $k = 80/9$ instead of 5, then computed $(80/9)^2 - 5$ . |

**Shortcut**



You never need to find  $x$  or  $y$  at all -- the entire problem collapses to one proportion ( $3k/20 = 3/4$ ) once the coefficient ratios are confirmed to already match.

### Desmos Tip

Substitute  $k = 5$  back into  $6x - 9y = 15$  and graph it alongside  $8x - 12y = 20$  (solved for  $y$ ) to confirm the two lines overlap exactly, verifying infinitely many solutions.

## Systems of Linear Equations — Problem 7 (Very Hard)

Let  $m$  equal the value of  $y$  in the solution to the system  $x + y = 15$  and  $x - y = 3$ . Using this value of  $m$ , for what value of  $n$  does the system  $3x + 4y = 7$  and  $mx + ny = -2$  have no solution?

**Formula / Rule Used:** Solve a preliminary system to fix an unknown coefficient, then apply the no-solution ratio condition  $A1/A2 = B1/B2$ , with  $C1/C2$  different, to the resulting system.

### Solution

**Step 1:** Solve the first system: add the two equations,  $(x + y) + (x - y) = 15 + 3$ , so  $2x = 18$  and  $x = 9$ .

**Step 2:** Subtract the two equations instead to get  $y$  directly:  $(x + y) - (x - y) = 15 - 3$ , so  $2y = 12$  and  $y = 6$ .

**Step 3:** So  $m = 6$ , the value of  $y$  from the first system.

**Step 4:** Substitute  $m = 6$  into the second system, which becomes  $3x + 4y = 7$  and  $6x + ny = -2$ .

**Step 5:** For this system to have no solution, the coefficient ratios must be equal:  $3/6 = 4/n$ .

**Step 6:** Simplify the left side:  $3/6 = 1/2$ , so  $1/2 = 4/n$ .

**Step 7:** Cross-multiply:  $n = 8$ .

**Step 8:** Confirm this is a true no-solution case (not infinitely many): the constant ratio is  $7/(-2) = -3.5$ , which does not equal  $1/2$ , so the lines are parallel but distinct.

**Answer:**  $n = 8$

### Tip

When a coefficient in a ratio-condition problem is defined as 'the value of' a specific variable from a separate system, fully solve that system first and clearly identify which variable ( $x$  or  $y$ ) supplies the coefficient before setting up any ratio.

### Trap

The two-stage setup tempts students to rush the first system and grab the wrong variable ( $x$  instead of  $y$ ) for  $m$ , or to mismatch which coefficient belongs in the numerator versus denominator when writing the ratio  $3/6 = 4/n$ , since the problem never explicitly labels which equation is 'first' and which is 'second'.

### Process of Elimination

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                                                                    |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $n = 12$              | Used the $x$ -value (9) from the first system as $m$ instead of the $y$ -value (6), then solved the ratio $3/9 = 4/n$ .                                                                                                                    |
| $n = 2$               | Correctly found $m = 6$ but inverted the coefficient ratio, comparing the second equation's $x$ -coefficient to the first's (6/3) instead of the first's to the second's (3/6), while keeping the $y$ -coefficients in the original order. |



|          |                                                                                                                                                                        |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $n = -8$ | Correctly found $m = 6$ but made a sign error setting up the ratio equation, writing $3/6 = -4/n$ instead of $3/6 = 4/n$ , which flips the sign of the resulting $n$ . |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Shortcut**

The first system's equations sum directly to  $2x = 18$  and subtract directly to  $2y = 12$ , so you can read off both  $x$  and  $y$  in one line each without full elimination setup -- useful for confirming  $m = 6$  quickly.

**Desmos Tip**

Graph  $x + y = 15$  and  $x - y = 3$  to confirm the intersection  $(9, 6)$ , then graph  $3x + 4y = 7$  and  $6x + 8y = -2$  (solved for  $y$ ) to visually confirm the lines are parallel and never intersect.

**Systems of Linear Equations — Problem 8 (Hard)**

**Solve the system  $3x + 4y = 7$  and  $2x - 3y = 16$ . What is the value of  $x - y$ ?**

**Formula / Rule Used:** Elimination by scaling both equations to a common coefficient, then compute the specific combination the question actually asks for.

**Solution**

**Step 1:** To eliminate  $y$ , find the LCM of 4 and 3, which is 12: multiply the first equation by 3 to get  $9x + 12y = 21$ .

**Step 2:** Multiply the second equation by 4 to get  $8x - 12y = 64$ .

**Step 3:** Add the two new equations:  $9x + 12y + 8x - 12y = 21 + 64$ , so  $17x = 85$ .

**Step 4:** Solve for  $x$ :  $x = 85/17 = 5$ .

**Step 5:** Substitute  $x = 5$  into  $2x - 3y = 16$ :  $10 - 3y = 16$ , so  $-3y = 6$  and  $y = -2$ .

**Step 6:** Compute the requested expression:  $x - y = 5 - (-2) = 5 + 2 = 7$ .

**Answer:**  $x - y = 7$

**Tip**

Once you solve for  $x$  and  $y$ , reread exactly what expression the question wants ( $x - y$ , not  $x$  alone, not  $y$  alone, not  $y - x$ ) before writing your final answer.

**Trap**

This problem is built so that  $x = 5$  and  $y = -2$  are both 'clean' numbers on their own, tempting a student to report just  $x = 5$  (answering 'what is  $x$ ' instead of 'what is  $x - y$ ') or to subtract in the wrong order -- either looks like a perfectly reasonable final answer if you stop reading carefully.

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------------------------------|
| 5                     | Solved correctly for $x = 5$ but reported it directly, forgetting the question asks for $x - y$ , not $x$ alone. |
| -2                    | Solved correctly for $y = -2$ but reported it directly, mistakenly treating $y$ as                               |



|           |                                                                                                                                               |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|
|           | the requested value instead of computing $x - y$ .                                                                                            |
| <b>-7</b> | Correctly found $x = 5$ and $y = -2$ but computed $y - x$ instead of $x - y$ , subtracting in the reversed order and getting -7 instead of 7. |

**Shortcut**

No shortcut beats solving fully here -- since the question asks for a specific combination ( $x - y$ ) rather than a simple sum or difference of the two original equations as given, you must find  $x$  and  $y$  individually before combining them.

**Desmos Tip**

Graph  $3x + 4y = 7$  and  $2x - 3y = 16$  in Desmos, read the intersection point  $(5, -2)$ , then compute  $5 - (-2) = 7$  directly from the coordinates.

### 3. Linear Inequalities

#### Linear Inequalities — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

| Term (English)                | Greek — Ελληνικά                  |
|-------------------------------|-----------------------------------|
| Inequality                    | Ανίσωση                           |
| Solution set                  | Σύνολο λύσεων                     |
| Compound inequality           | Σύνθετη ανίσωση                   |
| Flip the inequality sign      | Αναστροφή συμβόλου ανίσωσης       |
| Boundary line / shaded region | Οριακή ευθεία / σκιασμένη περιοχή |

#### Linear Inequalities — Condensed Theory

Inequalities are solved like equations, with one critical exception: multiplying or dividing by a negative number reverses the inequality sign.

##### The Flip Rule

**Multiply/divide by a negative → reverse the sign.**

Adding or subtracting (even negative numbers) never flips the sign — only multiplying or dividing by a negative does.

##### Compound (Three-Part) Inequalities

**$a < x < b$**

Apply the same operation to all three parts simultaneously; flip both signs if you multiply/divide by a negative.

##### Graphing in the xy-Plane

Solid line for  $\leq / \geq$  (boundary included); dashed line for  $< / >$  (boundary excluded). Shade above for  $y > / \geq$ , below for  $y < / \leq$ , or test a point like (0,0).

##### Systems of Inequalities

Graph each inequality; the solution is the overlapping shaded region. To check a point, substitute it into every inequality.

##### Word Problems

**'at least/no less than' →  $\geq$ . 'at most/no more than' →  $\leq$ .**



After solving, check whether the context requires an integer answer, and round down for a maximum or up for a minimum.

### Watch Out

Real-world quantities (people, tickets, boxes) often force you to round the algebraic answer to the nearest valid integer — always re-read what's being asked (max vs. min) before rounding.

## Linear Inequalities — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

### Linear Inequalities — Problem 1 (Hard)

The inequality  $3(x - 4) - 5x \geq k - 15$  has solution set  $x \leq 7$ . What is the value of  $k$ ?

**Formula / Rule Used:** Distribute and combine like terms, then divide by a negative number (which flips the inequality) to isolate  $x$ ; reverse-engineer an unknown constant from a given solution set.

#### Solution

**Step 1:** Expand the left side:  $3(x - 4) - 5x = 3x - 12 - 5x = -2x - 12$ , so the inequality becomes  $-2x - 12 \geq k - 15$ .

**Step 2:** Add 15 to both sides:  $-2x - 12 + 15 \geq k$ , which simplifies to  $-2x + 3 \geq k$ , or equivalently  $-2x \geq k - 3$ .

**Step 3:** Divide both sides by  $-2$  to isolate  $x$ . Dividing by a negative number reverses the inequality sign:  $x \leq (k - 3)/(-2)$ , which simplifies to  $x \leq (3 - k)/2$ .

**Step 4:** The problem states the solution set is  $x \leq 7$ , so set the boundary expressions equal:  $(3 - k)/2 = 7$ .

**Step 5:** Multiply both sides by 2:  $3 - k = 14$ .

**Step 6:** Solve for  $k$ :  $-k = 14 - 3 = 11$ , so  $k = -11$ .

**Step 7:** Check: substitute  $k = -11$  into the original inequality:  $3(x-4) - 5x \geq -11 - 15 = -26$ . The left side simplifies to  $-2x - 12$ , so  $-2x - 12 \geq -26$ , giving  $-2x \geq -14$ , and dividing by  $-2$  (flip) gives  $x \leq 7$ . Confirmed.

**Answer:  $k = -11$**

### Tip

When a solution set is given and you need to find a hidden constant, solve the inequality symbolically in terms of that constant first, then match the resulting boundary expression to the given boundary value.

### Trap

A student who forgets to flip the inequality sign when dividing by  $-2$  will set up the equation  $(k-3)/2 = 7$  instead of  $(3-k)/2 = 7$ , leading to the wrong value  $k = 11$  instead of the correct  $k = -11$ .

### Process of Elimination

**Tempting Wrong Answer**

**Exactly What Went Wrong**



|               |                                                                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>k = 11</b> | Divided $-2x \geq k - 3$ by $-2$ without reversing the inequality, setting up $(k-3)/2 = 7$ instead of $(3-k)/2 = 7$ , giving $k = 11$ instead of $-11$ .             |
| <b>k = -3</b> | Distributed $3(x-4)$ incorrectly as $3x - 4$ (forgetting to multiply the $-4$ by $3$ as well), which shifts the constant term and yields $k = -3$ instead of $-11$ .  |
| <b>k = 13</b> | Mis-copied the $-12$ term as $+12$ when isolating $k$ , writing $-2x + 12 \geq k - 15$ instead of $-2x - 12 \geq k - 15$ , which produces $k = 13$ instead of $-11$ . |

**Shortcut**

Skip solving the general inequality entirely: plug the boundary value  $x = 7$  directly into the original expression as an equality,  $-2(7) - 12 = k - 15$ , which gives  $k = -11$  in one line.

**Desmos Tip**

Graph  $y = -2x - 12$  and  $y = -26$  (using  $k = -11$ ) and confirm the graphs intersect at  $x = 7$ , matching the given boundary.

**Linear Inequalities — Problem 2 (Very Hard)**

Solve the compound inequality for  $x$ , and express the solution set using inequality notation:  $-3 < (2 - 5x)/3 \leq 4$ .

**Formula / Rule Used:** Compound (three-part) inequality: perform the same operation on all three parts, reversing both inequality signs and swapping the endpoint order when multiplying or dividing by a negative number.

**Solution**

**Step 1:** Multiply all three parts by  $3$  to clear the denominator:  $-3 \times 3 < 2 - 5x \leq 4 \times 3$ , giving  $-9 < 2 - 5x \leq 12$ .

**Step 2:** Subtract  $2$  from all three parts:  $-9 - 2 < -5x \leq 12 - 2$ , giving  $-11 < -5x \leq 10$ .

**Step 3:** Divide all three parts by  $-5$ . Since  $-5$  is negative, reverse both inequality signs and flip the order of the parts:  $-11/-5 > x \geq 10/-5$ .

**Step 4:** Simplify each side:  $-11/-5 = 11/5$  and  $10/-5 = -2$ , so the inequality reads  $11/5 > x \geq -2$ .

**Step 5:** Rewrite in standard increasing order (smallest to largest, left to right):  $-2 \leq x < 11/5$ .

**Step 6:** Check an endpoint: at  $x = -2$ ,  $(2 - 5(-2))/3 = (2 + 10)/3 = 12/3 = 4$ , which satisfies  $\leq 4$  with equality, confirming  $x = -2$  is correctly included.

**Step 7:** Check the other endpoint: at  $x = 11/5$ ,  $(2 - 5(11/5))/3 = (2 - 11)/3 = -9/3 = -3$ , which is NOT strictly greater than  $-3$  (it equals  $-3$ ), confirming  $x = 11/5$  is correctly excluded since the original inequality was strict ( $>$ ) on that side.

**Answer:**  $-2 \leq x < 11/5$

**Tip**

Clear the fraction first by multiplying every part by the denominator, then isolate  $x$  exactly as with a normal compound inequality, remembering to flip both inequality signs and reverse the order of the endpoints whenever you multiply or divide by a negative number.

**Trap**

A student who divides by -5 but forgets to also swap which endpoint is 'less than' and which is 'greater than' will report the answer as  $11/5 \leq x < -2$ , which is an empty (impossible) set since  $11/5$  is greater than  $-2$ .

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                                               |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $11/5 \leq x < -2$    | Flipped the inequality symbols correctly when dividing by -5 but forgot to also swap the left-right order of the two endpoints, producing an impossible set where the 'lower' bound is larger than the 'upper' bound. |
| $-2/5 \leq x < 1$     | Skipped clearing the fraction (never multiplied all three parts by 3) and instead subtracted 2 and divided by -5 directly on the un-cleared expression $(2 - 5x)/3$ , shifting every boundary value.                  |
| $-2 < x \leq 11/5$    | Correctly flipped the inequality direction and endpoint order but attached the strict '<' symbol to the wrong endpoint, swapping which side is open and which is closed.                                              |

**Shortcut**

No shortcut beats clearing the fraction first -- trying to divide by 3 termwise before combining like terms just recreates the same fraction later, so multiplying through immediately is already the fastest path.

**Desmos Tip**

Graph  $y = (2 - 5x)/3$  along with the horizontal lines  $y = -3$  and  $y = 4$ , then read off the x-interval where the slanted line stays between the two horizontal lines (open circle at  $x = 11/5$ , closed circle at  $x = -2$ ).

**Linear Inequalities — Problem 3 (Hard)**

A region R in the xy-plane is defined by the system of inequalities:  $y \leq -2x + 9$  and  $y > \frac{1}{3}x - 1$ . Determine which of the points P(3, 2), Q(0, 10), or S(5, -3) lies in region R.

**Formula / Rule Used:** A point satisfies a system of inequalities only if it satisfies every inequality in the system simultaneously -- substitute the point's coordinates directly into each inequality.

### Solution

**Step 1:** Test point P(3, 2) in the first inequality:  $2 \leq -2(3) + 9 = -6 + 9 = 3$ . Since  $2 \leq 3$  is true, P satisfies the first inequality.

**Step 2:** Test point P(3, 2) in the second inequality:  $2 > \frac{1}{3}(3) - 1 = 1 - 1 = 0$ . Since  $2 > 0$  is true, P satisfies the second inequality too, so P is in region R.

**Step 3:** Test point Q(0, 10) in the first inequality:  $10 \leq -2(0) + 9 = 9$ . Since  $10 \leq 9$  is false, Q fails the first inequality and is NOT in region R (no need to check the second inequality).

**Step 4:** Test point S(5, -3) in the first inequality:  $-3 \leq -2(5) + 9 = -10 + 9 = -1$ . Since  $-3 \leq -1$  is true, S passes the first inequality.

**Step 5:** Test point S(5, -3) in the second inequality:  $-3 > \frac{1}{3}(5) - 1 = \frac{5}{3} - 1 = \frac{2}{3}$ . Since  $-3 > \frac{2}{3}$  is false, S fails the second inequality, so S is NOT in region R.

**Step 6:** Only point P(3, 2) satisfies both inequalities simultaneously, so P is the point that lies in region R.

**Answer:** P(3, 2)

### Tip

Never rearrange or graph the inequalities to check a specific point -- just substitute its x- and y-coordinates directly into each inequality as given and evaluate; it's faster and avoids sign errors.

### Trap

A student who only checks the first inequality for each point (since both P and S pass it) might stop early and mistakenly accept S as a valid answer without testing the strict second inequality, missing that S fails  $y > \frac{1}{3}x - 1$ .

### Process of Elimination

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                                                                                   |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S(5, -3)              | Checked only the first inequality $y \leq -2x + 9$ for each point (which both P and S satisfy) and stopped there without testing the strict second inequality, incorrectly accepting S even though it fails $y > \frac{1}{3}x - 1$ .                      |
| Q(0, 10)              | Checked only the second inequality $y > \frac{1}{3}x - 1$ for each point -- Q passes it easily ( $10 > -1$ ) -- and picked Q without ever testing it against the first inequality, where it actually fails ( $10 \leq 9$ is false).                       |
| No point lies in R    | Misapplied the first inequality's substitution for P, computing $-2(3 + 9)$ instead of $-2(3) + 9$ (distributing across the sum instead of just the x-term), making P appear to fail the first inequality, so no point seemed to satisfy both conditions. |

### Shortcut



Since Q obviously fails the first inequality's y-value at  $x = 0$  (10 versus the boundary value 9), testing Q first eliminates it fastest, leaving only two candidates left to fully check.

### Desmos Tip

Graph both boundary lines  $y = -2x + 9$  (solid) and  $y = (1/3)x - 1$  (dashed) with the correct shading, then plot the three points to see visually that only P falls inside the overlapping shaded region.

## Linear Inequalities — Problem 4 (Hard)

A student committee in Rotterdam is chartering a bus for a class trip. The bus rental costs a flat 340 euros plus 12 euros per student who rides it. Each student pays a 25 euro activity fee to the committee. The committee also printed 80 information flyers for parents. The committee wants the total fees collected to be at least 500 euros more than the total bus rental cost. What is the minimum number of students needed for this to happen?

**Formula / Rule Used:** Translate 'at least' as  $\geq$ , set up a linear inequality in the unknown count, then round the solution to the correct integer in the direction that keeps the inequality true.

### Solution

**Step 1:** Let  $n$  represent the number of students riding the bus.

**Step 2:** Total fees collected by the committee:  $25n$ .

**Step 3:** Total bus rental cost:  $340 + 12n$ .

**Step 4:** Translate 'at least 500 euros more than the cost' into an inequality:  $25n \geq (340 + 12n) + 500$ .

**Step 5:** Simplify the right side:  $25n \geq 840 + 12n$ .

**Step 6:** Subtract  $12n$  from both sides:  $25n - 12n \geq 840$ , so  $13n \geq 840$ .

**Step 7:** Divide both sides by 13 (positive, no flip needed):  $n \geq 840/13 = 64.615\ldots$

**Step 8:** Since  $n$  must be a whole number of students, round UP to the next integer:  $n = 65$  (rounding down to 64 would give  $13(64) = 832$ , which is less than 840 and fails the requirement).

**Step 9:** Note that the flyer count (80) is never used in the calculation -- it is irrelevant to the fee-versus-cost relationship.

**Answer: 65 students**

### Tip

In minimum-value word problems with 'at least,' always solve the inequality first, then round the decimal boundary UP to the next whole number, since rounding down would fall short of the requirement.

### Trap

A student might round  $64.615\ldots$  down to 64 out of habit, or try to work the irrelevant flyer count of 80 into the equation -- explicitly verifying that  $n = 64$  gives  $832 < 840$  (a failure) shows why 65 is required.

### Process of Elimination

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                              |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 64 students           | Computed $n \geq 840/13 = 64.615\ldots$ but rounded down to the nearest whole number out of habit instead of up, submitting 64, which actually gives $13(64) = 832 < 840$ and fails the requirement. |



|                    |                                                                                                                                                                                                           |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>27 students</b> | Dropped the '500 euros more than' condition entirely, setting up $25n \geq 340 + 12n$ instead of $25n \geq 340 + 12n + 500$ , which gives $n = 27$ , far too low to satisfy the actual requirement.       |
| <b>23 students</b> | Combined the $12n$ term incorrectly by adding instead of subtracting when isolating $n$ , turning $25n - 12n \geq 840$ into $25n + 12n \geq 840$ , which gives $37n \geq 840$ and $n = 23$ instead of 65. |

**Shortcut**

Instead of expanding  $(340 + 12n) + 500$  fully, immediately combine the constants  $340 + 500 = 840$  mentally before writing the inequality, saving a simplification step.

**Desmos Tip**

Graph  $y = 25x$  and  $y = 840 + 12x$  and find where the first crosses above the second; any integer  $x$  at or to the right of that intersection point (rounded up) is a valid answer.

**Linear Inequalities — Problem 5 (Very Hard)**

**For what values of  $k$  is the slope of the line through the points  $(2, k)$  and  $(7, -3)$  less than  $-2$ ?**

**Formula / Rule Used:** Slope formula  $m = (y_2 - y_1)/(x_2 - x_1)$ , combined with solving a linear inequality where dividing by a negative number reverses the inequality.

**Solution**

**Step 1:** Write the slope formula using the two given points:  $m = (-3 - k)/(7 - 2) = (-3 - k)/5$ .

**Step 2:** Set up the inequality using the given condition that the slope is less than  $-2$ :  $(-3 - k)/5 < -2$ .

**Step 3:** Multiply both sides by 5 (positive, no flip needed):  $-3 - k < -10$ .

**Step 4:** Add 3 to both sides:  $-k < -10 + 3 = -7$ .

**Step 5:** Divide both sides by  $-1$  to isolate  $k$ . Since we're dividing by a negative number, reverse the inequality sign:  $k > 7$ .

**Step 6:** Check with a test value: let  $k = 8$ . Slope  $= (-3 - 8)/5 = -11/5 = -2.2$ , and  $-2.2 < -2$  is true, confirming  $k > 7$  is correct.

**Step 7:** Check the boundary: at  $k = 7$ , slope  $= (-3 - 7)/5 = -10/5 = -2$ , which is NOT less than  $-2$  (it's equal), confirming  $k = 7$  itself must be excluded, matching the strict inequality  $k > 7$ .

**Answer:**  $k > 7$

**Tip**

When slope is described using an inequality, first write the slope formula normally as an expression in terms of the unknown, then solve that expression as an ordinary inequality -- treat the slope formula as just another algebraic expression.

**Trap**

A student who mishandles the final divide-by  $-1$  step (forgetting to flip the sign) will conclude  $k < 7$  instead of the correct  $k > 7$  -- exactly backwards.

**Process of Elimination**



| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                                             |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $k < 7$               | Divided $-k < -7$ by $-1$ without reversing the inequality sign, concluding $k < 7$ instead of the correct $k > 7$ -- exactly backwards.                                                                            |
| $k < -13$             | Wrote the slope formula with the y-values subtracted in the wrong order relative to the x-values, using $(k + 3)/5$ instead of $(-3 - k)/5$ , which flips the sign of the whole expression and leads to $k < -13$ . |
| $k \geq 7$            | Solved the inequality correctly down to the boundary $k = 7$ but reported it as $k \geq 7$ (inclusive), forgetting the original slope condition was strict less-than, which excludes the boundary $k = 7$ itself.   |

**Shortcut**

Multiply the inequality  $(-3 - k)/5 < -2$  by 5 and then by  $-1$  in one combined move, immediately reversing to  $3 + k > 10$ , then solve  $k > 7$  directly without a separate 'add 3' step.

**Desmos Tip**

Graph  $y = (-3 - x)/5$  and the horizontal line  $y = -2$ , then observe the slanted graph lies below  $y = -2$  exactly when  $x > 7$ .

**Linear Inequalities — Problem 6 (Very Hard)**

A factory's quality-control rule states that a machined rod fails inspection if its length  $L$  (in centimeters) differs from the target length of 18 cm by 3 cm or more:  $|L - 18| \geq 3$ . The factory runs three shifts per day and produces 200 rods per shift. For what values of  $L$  does a rod PASS inspection? Express your answer as a compound inequality.

**Formula / Rule Used:** A rod passes inspection exactly when it does NOT satisfy the failing condition -- negate  $|L - 18| \geq 3$  to get  $|L - 18| < 3$ , then split the absolute value inequality into a three-part compound inequality.

**Solution**

**Step 1:** Passing is the logical opposite of failing. Since failing means  $|L - 18| \geq 3$ , passing means the negation:  $|L - 18| < 3$ .

**Step 2:** An absolute value inequality of the form  $|\text{expression}| < c$  (with  $c$  positive) splits into a compound inequality:  $-c < \text{expression} < c$ . Here that gives  $-3 < L - 18 < 3$ .

**Step 3:** Add 18 to all three parts of the compound inequality:  $-3 + 18 < L - 18 + 18 < 3 + 18$ .

**Step 4:** Simplify:  $15 < L < 21$ .

**Step 5:** Sanity check with the failing boundary: at  $L = 15$ ,  $|15 - 18| = |-3| = 3$ , which satisfies the ORIGINAL failing condition ( $\geq 3$ , since it equals 3), so  $L = 15$  should indeed fail and correctly be excluded from the passing range -- confirming the strict inequality ( $<$  not  $\leq$ ) is correct on both sides.

**Step 6:** Note that the production details (three shifts, 200 rods per shift) are not needed to answer the question about acceptable lengths.

**Answer:**  $15 < L < 21$  (centimeters)

**Tip**

When a problem defines a 'failing' or 'rejection' condition with  $\geq$  or  $>$ , remember that the 'passing' condition is its exact logical complement, which flips the inequality to  $<$  or  $\leq$  before you even start splitting the absolute value.

**Trap**

A student who solves  $|L - 18| \geq 3$  directly (getting  $L \leq 15$  OR  $L \geq 21$ , the FAILING range) and reports that as the answer to 'which rods pass' has answered the opposite question -- the passing range is the complement,  $15 < L < 21$ .

**Process of Elimination**

| Tempting Wrong Answer      | Exactly What Went Wrong                                                                                                                                                                                                      |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $L \leq 15$ or $L \geq 21$ | Solved the given failing condition $ L - 18  \geq 3$ directly and reported that FAILING range as the answer, instead of first negating it to find the PASSING condition $ L - 18  < 3$ .                                     |
| $15 \leq L \leq 21$        | Used $\leq$ instead of $<$ when splitting $ L - 18  < 3$ into $-3 < L - 18 < 3$ , incorrectly including the boundary values 15 and 21, which actually belong to the failing range since $ 15 - 18  = 3$ satisfies $\geq 3$ . |
| $12 < L < 24$              | Doubled the tolerance value (used 6 instead of 3) when splitting the absolute value inequality, apparently confusing the $\pm 3$ boundary with a full range width, giving $12 < L < 24$ instead of $15 < L < 21$ .           |

**Shortcut**

The boundary values (15 and 21) are identical whether you solve for the failing or passing condition, so you can solve whichever direction feels more natural (here, directly splitting  $|L - 18| < 3$ ) and just make sure the final inequality symbols match what 'passing' truly requires.

**Desmos Tip**

Graph  $y = \text{abs}(L - 18)$  and the horizontal line  $y = 3$ , then read off the L-interval where the V-shaped graph dips below the line -- that's the passing range  $15 < L < 21$ .

**Linear Inequalities — Problem 7 (Hard)**

**For how many integer values of  $x$  do both inequalities  $2x - 5 < 9$  and  $-3x + 4 \leq 19$  hold true at the same time?**

**Formula / Rule Used:** Solve each linear inequality separately, then take the intersection of the two solution sets before counting integer solutions.

**Solution**

**Step 1:** Solve the first inequality:  $2x - 5 < 9$ . Add 5 to both sides:  $2x < 14$ . Divide by 2 (positive, no flip):  $x < 7$ .

**Step 2:** Solve the second inequality:  $-3x + 4 \leq 19$ . Subtract 4 from both sides:  $-3x \leq 15$ . Divide by -3, which is negative, so reverse the inequality:  $x \geq -5$ .

**Step 3:** Combine both solution sets:  $x$  must satisfy  $x < 7$  AND  $x \geq -5$  at the same time, giving the intersection  $-5 \leq x < 7$ .

**Step 4:** List the integers in this range, including -5 but not including 7: -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6.

**Step 5:** Count the listed integers: there are 12 of them.

**Step 6:** Verify the endpoints: at  $x = -5$ ,  $-3(-5) + 4 = 19$ , which satisfies  $\leq 19$  (boundary correctly included); at  $x = 7$ ,  $2(7) - 5 = 9$ , which does NOT satisfy  $< 9$  (boundary correctly excluded).

**Answer: 12 integer values ( $x = -5$  through  $x = 6$ )**

**Tip**

When a problem asks how many integers satisfy two inequalities together, solve each one separately first, then intersect the two number lines before counting -- never try to merge the original inequalities into one expression.

**Trap**

A student who forgets to flip the inequality sign when dividing  $-3x \leq 15$  by -3, or who mishandles which boundary is included versus excluded, will miscount the integers in the intersection even though each inequality looks routine on its own.

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                         |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>13</b>             | Treated the strict inequality $2x - 5 < 9$ as if it were $\leq$ , including the boundary $x = 7$ , which actually fails the original strict inequality since $2(7) - 5 = 9$ is not less than 9. |
| <b>11</b>             | Treated $-3x + 4 \leq 19$ as if it were strictly $<$ , excluding the boundary $x = -5$ , which actually satisfies the original inequality with equality since $-3(-5) + 4 = 19$ .               |
| <b>14</b>             | Added 4 to both sides instead of subtracting it when isolating $-3x$ , turning $-3x \leq 15$ into $-3x \leq 23$ , which shifts the lower bound down and produces an incorrect count of 14.      |

**Shortcut**

Since both inequalities solve cleanly to integer boundaries, skip listing every integer individually -- just compute the upper bound minus the lower bound,  $7 - (-5) = 12$ , which directly gives the count since the lower bound is included and the upper bound is excluded.

**Desmos Tip**

Graph  $y = 2x - 5$  with the line  $y = 9$ , and  $y = -3x + 4$  with the line  $y = 19$ , to visually confirm the two boundary  $x$ -values  $-5$  and  $7$  before counting the integers between them.

**Linear Inequalities — Problem 8 (Very Hard)**

**For a positive constant  $a$ , the inequality  $a|x - 5| \geq 12$  has solution set  $x \leq -1$  or  $x \geq 11$ . What is the value of  $a$ ?**

**Formula / Rule Used:** Splitting an absolute value inequality of the form  $|\text{expression}| \geq c$  into two linear branches, combined with reverse-engineering an unknown coefficient from a given solution set.

**Solution**

**Step 1:** Divide both sides by  $a$ . Since  $a$  is positive, the inequality direction stays the same:  $|x - 5| \geq 12/a$ .

**Step 2:** Split the absolute value inequality into its two branches:  $x - 5 \geq 12/a$ , or  $x - 5 \leq -12/a$ .

**Step 3:** Solve each branch for  $x$ :  $x \geq 5 + 12/a$ , or  $x \leq 5 - 12/a$ .

**Step 4:** Match the first branch to the given boundary  $x \geq 11$ :  $5 + 12/a = 11$ .

**Step 5:** Solve for  $a$ :  $12/a = 11 - 5 = 6$ , so  $a = 12/6 = 2$ .

**Step 6:** Verify with the second branch using  $a = 2$ :  $5 - 12/2 = 5 - 6 = -1$ , which exactly matches the given boundary  $x \leq -1$ , confirming  $a = 2$  works for both branches.

**Step 7:** Since the problem specifies  $a$  is a positive constant,  $a = 2$  satisfies that condition as well.

**Answer:  $a = 2$**

**Tip**

When an unknown coefficient sits inside an absolute value inequality and you're given the final solution set, split the absolute value into its two linear branches symbolically first, then match either branch's boundary to the corresponding given boundary value to solve for the unknown.

**Trap**

A student who solves  $5 + 12/a = 11$  by adding 5 to both sides instead of subtracting it will get  $12/a = 16$  instead of  $12/a = 6$ , producing  $a = 3/4$  instead of the correct  $a = 2$ .

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $a = 3/4$             | Solved $5 + 12/a = 11$ by adding 5 to both sides instead of subtracting it, getting $12/a = 16$ instead of $12/a = 6$ , which gives $a = 3/4$ instead of $a = 2$ .                                                                                            |
| $a = 6$               | Mishandled isolating the absolute value by subtracting $a$ from 12 instead of dividing by $a$ , turning $ x - 5  \geq 12/a$ into $ x - 5  \geq 12 - a$ , which leads to $a = 6$ instead of $a = 2$ .                                                          |
| $a = -2$              | Dropped the negative sign on the second branch (wrote $x - 5 \leq 12/a$ instead of $x - 5 \leq -12/a$ ), then matched $5 + 12/a$ to the boundary $-1$ instead of 11, producing $a = -2$ , which also violates the problem's requirement that $a$ be positive. |



### Shortcut

Once you split into the two branches  $5 + 12/a$  and  $5 - 12/a$ , you only need to match ONE branch to its corresponding boundary (say  $5 + 12/a = 11$ ) to solve for  $a$  -- the other branch is just a free check, not a second equation to solve.

### Desmos Tip

Graph  $y = 2 \cdot \text{abs}(x - 5)$  (using the solved value  $a = 2$ ) together with the horizontal line  $y = 12$ , then confirm the graph crosses  $y = 12$  exactly at  $x = -1$  and  $x = 11$ , matching the given solution set.

## 4. Absolute Value

### Absolute Value — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

| Term (English)                | Greek — Ελληνικά                  |
|-------------------------------|-----------------------------------|
| Absolute value                | Απόλυτη τιμή                      |
| Distance from zero            | Απόσταση από το μηδέν             |
| Two cases (positive/negative) | Δύο περιπτώσεις (θετική/αρνητική) |
| Extraneous solution           | Ξένη (μη αποδεκτή) λύση           |

### Absolute Value — Condensed Theory

Absolute value is distance from zero and is always non-negative. Every absolute value equation or inequality splits into two cases.

#### Solving $|X| = a$

**If  $a \geq 0$ :  $X = a$  or  $X = -a$ . If  $a < 0$ : no solution.**

Isolate the absolute value first, then split into the two cases.

#### Absolute Value Inequalities

**$|X| < a \rightarrow -a < X < a$ .  $|X| > a \rightarrow X < -a$  or  $X > a$  ( $a > 0$ )**

'Less than' becomes a compound (between) inequality; 'greater than' becomes two separate branches.

#### $|A| = |B|$

**$A = B$  or  $A = -B$**

Solve both resulting equations and check each solution in the original equation.

#### Extraneous Solutions

Whenever a variable appears both inside and outside the absolute value bars (or gets squared), you must check every solution in the ORIGINAL equation — one branch often fails.

#### Distance Interpretation

**$|x - a| = \text{distance between } x \text{ and } a$**

Useful for tolerance / 'within so many units of a target' word problems:  $|\text{actual} - \text{target}| \leq \text{tolerance}$ .

**⚠ Watch Out**

If isolating the absolute value ever produces  $|X| = (\text{negative number})$ , stop — there is no solution. Don't split into cases anyway.

**Absolute Value — 8 Worked Problems**

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

**Absolute Value — Problem 1 (Hard)**

Solve for  $x$ :  $|2x - 5| = 3x - 4$ . Check for extraneous solutions and give only the valid solution(s).

**Formula / Rule Used:**  $|X| = a$  splits into  $X = a$  or  $X = -a$ ; any solution must be checked in the original equation.

**Solution**

**Step 1:** Since the variable appears both inside and outside the absolute value bars, split into two cases.

**Step 2:** Case 1:  $2x - 5 = 3x - 4$ . Solve:  $2x - 3x = -4 + 5$ , so  $-x = 1$ , giving  $x = -1$ .

**Step 3:** Case 2:  $2x - 5 = -(3x - 4) = -3x + 4$ . Solve:  $2x + 3x = 4 + 5$ , so  $5x = 9$ , giving  $x = 9/5$ .

**Step 4:** Check  $x = -1$  in the original equation: left side =  $|2(-1) - 5| = |-7| = 7$ , right side =  $3(-1) - 4 = -7$ . Since 7 does not equal -7,  $x = -1$  is extraneous and must be rejected.

**Step 5:** Check  $x = 9/5$ : left side =  $|2(9/5) - 5| = |18/5 - 25/5| = |-7/5| = 7/5$ , right side =  $3(9/5) - 4 = 27/5 - 20/5 = 7/5$ . These match.

**Step 6:** The only valid solution is  $x = 9/5$ .

**Answer:**  $x = 9/5$

**Tip**

Whenever the variable also sits outside the absolute value bars, treat both algebraic results as candidates only — substitute each back into the ORIGINAL equation before accepting it.

**Trap**

A rushed student stops after solving the two linear cases and reports  $x = -1$  as a valid answer too, never noticing it forces the absolute value expression to equal a negative number (-7), which is impossible.

**Process of Elimination**

| Tempting Wrong Answer  | Exactly What Went Wrong                                                                                                                                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $x = -1$               | Solved Case 1 correctly to $x = -1$ but never substituted it back into the original equation, so did not catch that it makes the right side negative $(3(-1) - 4 = -7)$ , which an absolute value can never equal. |
| $x = 1/5$              | When expanding Case 2's negative sign, wrote $-(3x - 4)$ as $-3x - 4$ instead of correctly distributing to $-3x + 4$ , producing the wrong linear equation $2x - 5 = -3x - 4$ and the wrong root.                  |
| $x = -1$ and $x = 9/5$ | Listed both algebraic candidates as final answers, treating the extraneous-                                                                                                                                        |



|  |                                                                                              |
|--|----------------------------------------------------------------------------------------------|
|  | solution check as optional instead of mandatory whenever the variable sits outside the bars. |
|--|----------------------------------------------------------------------------------------------|

**Shortcut**

Before checking by substitution, you can flag a candidate as suspicious instantly by testing whether the bare side ( $3x-4$ ) is negative at that  $x$ -value, since an absolute value can never equal a negative number — this marks  $x=-1$  as doubtful before any arithmetic.

**Desmos Tip**

Graph  $y = \text{abs}(2x-5)$  and  $y = 3x-4$  and look at the actual intersection point — Desmos shows only one crossing, near  $x = 1.8$ , confirming  $x = -1$  is not a real solution.

**Absolute Value — Problem 2 (Hard)**

Solve for all values of  $x$ :  $5 + |2x - 6| = |x + 3| + 5$ .

**Formula / Rule Used:**  $|A| = |B|$  means  $A = B$  or  $A = -B$ .

**Solution**

**Step 1:** Subtract 5 from both sides to strip away the disguise:  $|2x - 6| = |x + 3|$ .

**Step 2:** Since  $|A| = |B|$  means  $A = B$  or  $A = -B$ , set up both cases using  $A = 2x - 6$  and  $B = x + 3$ .

**Step 3:** Case 1:  $2x - 6 = x + 3$ . Solve:  $2x - x = 3 + 6$ , so  $x = 9$ .

**Step 4:** Case 2:  $2x - 6 = -(x + 3) = -x - 3$ . Solve:  $2x + x = -3 + 6$ , so  $3x = 3$ , giving  $x = 1$ .

**Step 5:** Verify  $x = 9$ :  $|2(9)-6| = |12| = 12$  and  $|9+3| = |12| = 12$ , so both sides of the simplified equation match.

**Step 6:** Verify  $x = 1$ :  $|2(1)-6| = |-4| = 4$  and  $|1+3| = |4| = 4$ , so both sides match here too.

**Step 7:** Both candidates check out, so the solutions are  $x = 1$  and  $x = 9$ .

**Answer:**  $x = 1$  and  $x = 9$

**Tip**

First strip away anything added or subtracted outside the bars until the equation is purely  $|A| = |B|$ , then split into  $A = B$  and  $A = -B$ .

**Trap**

A rushed student splits into cases before simplifying, writing ' $2x-6+5 = x+3$ ' and ' $2x-6+5 = -(x+3)$ ' by dragging the extra  $+5$  into the case equations, which produces the wrong  $x$ -values entirely.

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $x = 4$               | Incorrectly folded the $+5$ into the bars, solving $2x-6+5 = x+3$ (as if $5+ 2x-6 $ meant $ 2x-6+5 $ ) instead of first subtracting 5 from both sides to isolate $ 2x-6  =  x+3 $ . |
| $x = -2/3$            | Made the same folding-in error but from the negative-case version,                                                                                                                  |



|         |                                                                                                                                                          |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|         | solving $2x-6+5 = -(x+3)$ instead of the correct Case 2 equation $2x-6 = -(x+3)$ .                                                                       |
| $x = 9$ | Solved only Case 1 ( $2x-6 = x+3$ ) and never set up Case 2 ( $2x-6 = -(x+3)$ ), so reported one solution and missed the second valid solution $x = 1$ . |

**Shortcut**

Once simplified to  $|2x-6| = |x+3|$ , you can solve  $2x - 6 = \text{plus-or-minus}(x+3)$  as one combined step instead of writing two fully separate case equations.

**Desmos Tip**

Graph  $y = 5 + \text{abs}(2x-6)$  and  $y = \text{abs}(x+3) + 5$  and read off both intersection x-coordinates directly, which confirms  $x = 1$  and  $x = 9$  without any algebra.

**Absolute Value — Problem 3** (Very Hard)

The inequality  $5|x - 3| - 7 > k$  has solution set  $x < -1$  or  $x > 7$ . What is the value of  $k$ ?

**Formula / Rule Used:**  $|X| > a$  ( $a > 0$ ) splits into  $X > a$  or  $X < -a$ .

**Solution**

**Step 1:** Isolate the absolute value expression first: add 7 to both sides to get  $5|x-3| > k + 7$ .

**Step 2:** Divide both sides by 5:  $|x - 3| > (k+7)/5$ .

**Step 3:** An inequality of the form  $|X| > a$  splits into  $X > a$  or  $X < -a$ , so  $x - 3 > (k+7)/5$  or  $x - 3 < -(k+7)/5$ .

**Step 4:** Rewrite each branch by adding 3:  $x > 3 + (k+7)/5$ , or  $x < 3 - (k+7)/5$ .

**Step 5:** Match the given upper boundary  $x > 7$  to the first branch:  $3 + (k+7)/5 = 7$ , so  $(k+7)/5 = 4$ , giving  $k + 7 = 20$ , so  $k = 13$ .

**Step 6:** Check against the given lower boundary  $x < -1$ :  $3 - (k+7)/5 = -1$ , so  $(k+7)/5 = 4$  again, giving the same  $k = 13$ .

**Step 7:** Both boundaries agree, so  $k = 13$ .

**Answer:**  $k = 13$

**Tip**

When you're given the finished solution set of an absolute value inequality and asked for a hidden constant, rebuild the general  $|x - \text{center}| > \text{distance}$  form and match its two boundary numbers to the given ones algebraically.

**Trap**

A rushed student plugs  $x = 7$  or  $x = -1$  directly into the original inequality as if it were an equation, but forgets to first isolate  $|x-3|$  by undoing the  $-7$  and dividing by 5, and often reports  $k = 7$  by ignoring the coefficient of 5 entirely.

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                          |
|-----------------------|----------------------------------------------------------------------------------|
| $k = 20$              | Dropped the $-7$ when isolating, treating $5 x-3 -7 > k$ as $5 x-3  > k$ , which |



|             |                                                                                                                                                          |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | gives $ x-3  > k/5$ ; matching the boundary $x=7$ then gives $k/5 = 4$ , so $k = 20$ instead of 13.                                                      |
| $k = -3$    | Correctly isolated the absolute value but forgot to divide by 5, using $3 + (k+7) = 7$ instead of $3 + (k+7)/5 = 7$ , giving $k = -3$ .                  |
| $k = -31/5$ | When solving $(k+7)/5 = 4$ , cross-multiplied backwards, writing $k+7 = 4/5$ instead of $k+7 = 4(5) = 20$ , producing a wrong fractional value for $k$ . |

**Shortcut**

Since 3 is exactly the midpoint of the two boundary values -1 and 7, the 'radius' distance is  $(7 - (-1))/2 = 4$ , so  $(k+7)/5 = 4$  can be written immediately without deriving both boundary equations separately.

**Desmos Tip**

Graph  $y = 5 \cdot \text{abs}(x-3) - 7$  and the horizontal line  $y = k$ , then adjust  $k$  until the shaded region's edges land exactly at  $x = -1$  and  $x = 7$  to visually confirm  $k = 13$ .

**Absolute Value — Problem 4 (Hard)**

A quality-control station in Patras accepts a 500 mL bottle only if its actual filled volume  $V$  satisfies  $|V - 500| \leq 4.5$ . One bottle was rejected because its measured volume was outside the acceptable tolerance by exactly 1.8 mL on the low side. What was that bottle's measured volume?

**Formula / Rule Used:** Tolerance statements translate to  $|\text{actual} - \text{target}| \leq \text{tolerance}$ , which unfolds into a compound inequality.

**Solution**

**Step 1:** Translate the tolerance rule into an inequality:  $|V - 500| \leq 4.5$ .

**Step 2:** Rewrite this without absolute value bars as a compound inequality:  $-4.5 \leq V - 500 \leq 4.5$ .

**Step 3:** Add 500 to all three parts:  $495.5 \leq V \leq 504.5$ , so any volume from 495.5 mL to 504.5 mL passes inspection.

**Step 4:** The bottle was rejected on the low side, meaning its measured volume fell below the minimum acceptable bound of 495.5 mL.

**Step 5:** 'Outside the tolerance by exactly 1.8 mL' means the measured volume was 1.8 mL less than that lower bound of 495.5 mL, not 1.8 mL less than the target of 500 mL.

**Step 6:** Compute:  $V = 495.5 - 1.8 = 493.7$  mL.

**Step 7:** The bottle's measured volume was 493.7 mL.

**Answer:**  $V = 493.7$  mL

**Tip**

Translate a tolerance statement 'within  $c$  of a target  $T$ ' as  $|\text{actual} - T| \leq c$ , then convert it into a normal compound inequality before doing any arithmetic with the boundary values.

**Trap**



A rushed student subtracts 1.8 directly from the target 500 (getting 498.2) instead of from the lower boundary 495.5, forgetting that the bottle was already outside the allowed window before being pushed 1.8 mL further away.

### Process of Elimination

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>V = 498.2 mL</b>   | Subtracted 1.8 from the target 500 directly instead of from the lower tolerance bound 495.5, ignoring that the bottle was already at the edge of the accepted window before failing further. |
| <b>V = 506.3 mL</b>   | Added 1.8 to the upper bound 504.5 instead of subtracting from the lower bound, misreading 'low side' as meaning the upper boundary was the one exceeded.                                    |
| <b>V = 494.2 mL</b>   | Mis-copied the tolerance value as 4 mL instead of 4.5 mL, computing the lower bound as $500 - 4 = 496$ , then subtracting 1.8 to get 494.2 mL instead of the correct 493.7 mL.               |

### Shortcut

Since the failure happened only on the low side, you can skip finding the upper bound (504.5) entirely and work with just the lower bound  $500 - 4.5 = 495.5$ , saving a step.

### Desmos Tip

Type  $\text{abs}(V-500) \leq 4.5$  into Desmos as a one-variable inequality to see the accepted interval  $[495.5, 504.5]$  plotted on the number line, then confirm 493.7 falls outside it on the left.

A tutor in Athens charges 35 euros per hour and is currently reviewing the equation  $|4x + 1| = x - 6$  with a student during a 50-minute session. Solve the equation for  $x$ , or determine that no real solution exists.

**Formula / Rule Used:**  $|X|$  can never equal a negative number, so any candidate solution must make the bare side non-negative.

### Solution

**Step 1:** Ignore the tutoring and pricing details — they don't affect the algebra. Split the equation into two cases as usual.

**Step 2:** Case 1:  $4x + 1 = x - 6$ . Solve:  $4x - x = -6 - 1$ , so  $3x = -7$ , giving  $x = -7/3$ .

**Step 3:** Case 2:  $4x + 1 = -(x - 6) = -x + 6$ . Solve:  $4x + x = 6 - 1$ , so  $5x = 5$ , giving  $x = 1$ .

**Step 4:** Before accepting either value, note that  $|4x+1|$  is never negative, so any true solution requires the right side  $x - 6$  to be greater than or equal to 0, meaning  $x \geq 6$ .

**Step 5:** Check  $x = -7/3$ : this does not satisfy  $x \geq 6$ . Directly verify:  $|4(-7/3)+1| = |-28/3+3/3| = |-25/3| = 25/3$ , but  $x - 6 = -7/3 - 18/3 = -25/3$ . Since  $25/3$  does not equal  $-25/3$ , this is extraneous.

**Step 6:** Check  $x = 1$ : this also fails  $x \geq 6$ . Directly verify:  $|4(1)+1| = 5$ , but  $x - 6 = -5$ . Since 5 does not equal  $-5$ , this is extraneous too.

**Step 7:** Both algebraic candidates fail the check, so the equation has no real solution.

**Answer:** No solution

### Tip

Before trusting either algebraic case in an equation like  $|\text{expr}| = (\text{linear expression})$ , check whether that linear expression can even be non-negative at the candidate  $x$ -value — an absolute value can never equal a negative number.

### Trap

A rushed student solves both linear cases, gets  $x = -7/3$  and  $x = 1$ , and reports one or both as answers without ever substituting back into the original equation, missing that neither actually works.

### Process of Elimination

| Tempting Wrong Answer  | Exactly What Went Wrong                                                                                                                                                                              |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $x = -7/3$             | Solved Case 1 ( $4x+1 = x-6$ ) correctly but never checked whether $x-6$ could be negative, submitting the algebraic root without noticing $ 4x+1 $ cannot equal the resulting negative value there. |
| $x = 1$                | Solved Case 2 ( $4x+1 = -(x-6)$ ) correctly but likewise never verified the result against the original equation, missing that it also forces the right side to be negative.                         |
| $x = -7/3$ and $x = 1$ | Listed both algebraic case solutions as the final answer without performing the required non-negativity check on either one, treating extraneous-solution verification as unnecessary.               |

### Shortcut



You can often spot 'no solution' quickly by testing only the larger candidate against the requirement  $x \geq 6$  first — since even  $x = 1$  already fails, both branches are immediately suspect without doing the full substitution check on each.

### Desmos Tip

Graph  $y = \text{abs}(4x+1)$  and  $y = x - 6$  — the two curves never intersect anywhere, visually confirming there is no solution.

## Absolute Value — Problem 6 (Very Hard)

Let  $m$  be the larger solution to the equation  $|3m - 5| = 2m + 1$ . Then determine the number of real solutions of the quadratic equation  $x^2 + 4x + (m - 9) = 0$  by evaluating its discriminant.

**Formula / Rule Used:** Solve  $|X| = a$  in two cases first; then use the discriminant  $b^2 - 4ac$  to count real roots of a quadratic.

### Solution

**Step 1:** Solve  $|3m - 5| = 2m + 1$  by splitting into two cases.

**Step 2:** Case 1:  $3m - 5 = 2m + 1$ . Solve:  $3m - 2m = 1 + 5$ , so  $m = 6$ . Check:  $|3(6) - 5| = |13| = 13$ , and  $2(6) + 1 = 13$ . Valid.

**Step 3:** Case 2:  $3m - 5 = -(2m + 1) = -2m - 1$ . Solve:  $3m + 2m = -1 + 5$ , so  $5m = 4$ , giving  $m = 4/5$ . Check:  $|3(4/5) - 5| = |12/5 - 25/5| = |-13/5| = 13/5$ , and  $2(4/5) + 1 = 8/5 + 5/5 = 13/5$ . Also valid.

**Step 4:** Both  $m = 6$  and  $m = 4/5$  satisfy the original equation, but the problem asks for the larger one, so  $m = 6$ .

**Step 5:** Substitute  $m = 6$  into the quadratic:  $x^2 + 4x + (6 - 9) = 0$ , which simplifies to  $x^2 + 4x - 3 = 0$ .

**Step 6:** Compute the discriminant:  $b^2 - 4ac = 4^2 - 4(1)(-3) = 16 + 12 = 28$ .

**Step 7:** Since the discriminant 28 is positive, the quadratic has two distinct real solutions.

**Answer:** Two distinct real solutions ( $m = 6$ , discriminant = 28)

### Tip

When a problem chains an absolute value equation into a second calculation, fully solve and verify the absolute value part first — including checking every candidate — before touching the second concept.

### Trap

A rushed student uses whichever solution of the absolute value equation they find first (often  $m = 4/5$  from Case 2) instead of comparing both valid solutions and correctly selecting the larger one as the problem instructs.

### Process of Elimination

| Tempting Wrong Answer                                          | Exactly What Went Wrong                                                                                                                                                                                    |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Two distinct real solutions ( $m = 4/5$ , discriminant = 48.8) | Used the smaller root $m = 4/5$ from Case 2 instead of correctly selecting the larger solution $m = 6$ as the problem requires, propagating the wrong $m$ into the discriminant.                           |
| Two distinct real solutions (discriminant = 4)                 | Substituted $m = 6$ correctly but made a sign error simplifying the constant term, writing $9 - m = 3$ instead of $m - 9 = -3$ , giving the wrong quadratic $x^2 + 4x + 3 = 0$ and the wrong discriminant. |

**Two distinct real solutions  
(discriminant = 16)**

Forgot to square  $b$  in the discriminant formula, using  $b = 4$  directly instead of  $b^2 = 16$ , computing  $4 - 4(1)(-3) = 16$  instead of the correct  $16 - 4(1)(-3) = 28$ .

**Shortcut**

Since both cases of  $|3m-5| = 2m+1$  check out here, there's no extraneous-root elimination needed — you can go straight to comparing 6 and  $4/5$  and move on to the quadratic without re-verifying anything twice.

**Desmos Tip**

Graph  $y = \text{abs}(3m-5)$  and  $y = 2m+1$  to instantly read off both intersection  $m$ -values (6 and 0.8), then graph  $y = x^2+4x-3$  separately to see it crosses the  $x$ -axis twice, confirming two real solutions.

**Absolute Value — Problem 7 (Very Hard)**

Find all real solutions to the equation  $||x - 4| - 7| = 3$ .

**Formula / Rule Used:** Nested absolute value: apply the  $|X| = a$  rule to the outer bars first ( $X = a$  or  $X = -a$ ), then apply the same rule again to each resulting inner absolute value equation.

**Solution**

**Step 1:** Let  $X = |x - 4| - 7$ , so the equation is  $|X| = 3$ . This splits into two outer cases:  $X = 3$  or  $X = -3$ .

**Step 2:** Outer Case A:  $|x - 4| - 7 = 3$ , so  $|x - 4| = 10$ .

**Step 3:** Split Outer Case A into its own two inner cases:  $x - 4 = 10$ , giving  $x = 14$ , or  $x - 4 = -10$ , giving  $x = -6$ .

**Step 4:** Outer Case B:  $|x - 4| - 7 = -3$ , so  $|x - 4| = 4$ .

**Step 5:** Split Outer Case B into its own two inner cases:  $x - 4 = 4$ , giving  $x = 8$ , or  $x - 4 = -4$ , giving  $x = 0$ .

**Step 6:** Since the outer equation's right side (3) is a positive constant, both outer cases are automatically valid, so no candidate needs to be discarded here.

**Step 7:** Verify one candidate,  $x = 0$ :  $|0 - 4| = 4$ , then  $|4 - 7| = |-3| = 3$ . Matches the original equation.

**Step 8:** The four real solutions are  $x = -6, 0, 8$ , and  $14$ .

**Answer:**  $x = -6, 0, 8, 14$

**Tip**

When you see an absolute value nested inside another absolute value, work from the outside in — first split the outer  $|X| = a$  into  $X = a$  and  $X = -a$  as two full equations, then split each of those inner absolute value equations again.

**Trap**

The most common failure is treating this as a single-layer absolute value problem and solving only  $|x-4| - 7 = 3$ , never realizing that  $|x-4| - 7 = -3$  is an equally valid second outer case, which silently discards two of the four correct solutions.

**Process of Elimination**

**Tempting Wrong Answer**

**Exactly What Went Wrong**



|              |                                                                                                                                                                                                                  |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $x = -6, 14$ | Solved only the outer case $ x-4  - 7 = 3$ , forgetting to also set up $ x-4  - 7 = -3$ , so missed the two solutions (0 and 8) that come from the second outer case.                                            |
| $x = 0, 8$   | Solved only the outer case $ x-4  - 7 = -3$ , forgetting the other outer case $ x-4  - 7 = 3$ , so missed the two solutions (-6 and 14) that come from the first outer case.                                     |
| $x = 8, 14$  | For both inner equations $ x-4  = 10$ and $ x-4  = 4$ , took only the positive branch ( $x-4=10$ and $x-4=4$ ) and never solved the negative branch ( $x-4=-10$ and $x-4=-4$ ), so missed $x = -6$ and $x = 0$ . |

**Shortcut**

Once you find  $|x-4| = 10$  and  $|x-4| = 4$ , you can write all four x-values immediately as  $4 \pm 10$  and  $4 \pm 4$  without redoing separate case equations for each.

**Desmos Tip**

Graph  $y = \text{abs}(\text{abs}(x-4)-7)$  and  $y = 3$ , then read off the four x-coordinates where they intersect: -6, 0, 8, and 14.

**Absolute Value — Problem 8** (Very Hard)

For all real numbers  $x$ , what is the minimum possible value of the expression  $|2x - 10| + |2x + 4|$ ?

**Formula / Rule Used:**  $|x-a| + |x-b|$  represents the total distance from  $x$  to two fixed points  $a$  and  $b$  on the number line; its minimum equals  $|a-b|$ , attained for any  $x$  between them. Factor out a shared coefficient before applying this rule.

**Solution**

**Step 1:** Factor the coefficient 2 out of each absolute value term:  $|2x - 10| = |2(x-5)| = 2|x-5|$ , and  $|2x + 4| = |2(x+2)| = 2|x+2|$ .

**Step 2:** The expression becomes  $2|x-5| + 2|x+2| = 2(|x-5| + |x+2|)$ .

**Step 3:**  $|x-5| + |x+2|$  is the sum of the distances from  $x$  to the two fixed points 5 and -2 on the number line.

**Step 4:** This sum of distances is minimized whenever  $x$  lies anywhere between -2 and 5 (inclusive), and the minimum value equals the distance between the two points:  $5 - (-2) = 7$ .

**Step 5:** Multiply by the factored-out coefficient of 2: minimum value =  $2 \times 7 = 14$ .

**Step 6:** Confirm with a test point inside the interval, say  $x = 0$ :  $|2(0)-10| + |2(0)+4| = |-10| + |4| = 10 + 4 = 14$ , matching the predicted minimum.

**Answer:** 14

**Tip**

When a sum of two absolute values shares a common coefficient, factor it out first, then recognize  $|x-a| + |x-b|$  as the total distance from  $x$  to two fixed points — a sum that is minimized between them and equals the distance between the points.

**Trap**

A rushed student forgets to factor out the shared coefficient of 2 and reports the raw anchor-point distance  $|5 - (-2)| = 7$  as the final answer, never doubling it back to account for the coefficient.

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7                     | Forgot to factor out and reapply the shared coefficient of 2 from each term, computing only the anchor-point distance $ 5 - (-2)  = 7$ and never multiplying it back by 2.                                                |
| 18                    | Incorrectly factored $ 2x+4 $ as $2 x+4 $ instead of $2 x+2 $ (a slip in the factoring step), which shifted the second anchor point to -4 and gave $2 \times  5 - (-4)  = 2 \times 9 = 18$ instead of $2 \times 7 = 14$ . |
| 6                     | Added the two anchor points ( $5 + (-2) = 3$ ) instead of finding the distance between them ( $5 - (-2) = 7$ ), then multiplied by 2 to get 6 instead of 14.                                                              |

**Shortcut**

Once you factor out the shared coefficient, no calculus or test points are needed — the minimum of  $|x-a| + |x-b|$  is always just the distance between  $a$  and  $b$ , so  $2 \times 7 = 14$  can be written in one line.

**Desmos Tip**

Graph  $y = \text{abs}(2x-10) + \text{abs}(2x+4)$  and trace the flat bottom of the V-shaped graph between  $x = -2$  and  $x = 5$ , where  $y$  stays constant at 14, confirming the minimum.

## 5. Exponent Rules

### Exponent Rules — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

| Term (English)    | Greek — Ελληνικά   |
|-------------------|--------------------|
| Exponent (power)  | Εκθέτης (δύναμη)   |
| Base              | Βάση               |
| Exponent rules    | Ιδιότητες δυνάμεων |
| Negative exponent | Αρνητικός εκθέτης  |
| Common base       | Κοινή βάση         |

### Exponent Rules — Condensed Theory

Every exponent rule follows from repeated multiplication. The SAT tests both simplifying exponential expressions and solving exponential equations.

#### Product & Quotient Rules

Same base: add exponents when multiplying, subtract when dividing.

#### Power Rules

Raising a power to a power multiplies exponents; a product or quotient raised to a power distributes to each factor.

#### Zero & Negative Exponents

A negative exponent means 'reciprocal,' never a negative number.

#### Fractional Exponents

The denominator of the exponent is the root; the numerator is the power.

#### Solving Exponential Equations

If (same base), then  $m = n$ .



If the bases differ but share a common factor, rewrite both sides with that common base first, then set exponents equal.

### Solving Fractional-Exponent Equations

If  $a^{\frac{m}{n}} = b^{\frac{p}{q}}$ , then  $a^{\frac{mq}{nq}} = b^{\frac{pn}{qn}}$ .

Raise both sides to the reciprocal power to isolate  $x$ .



### Watch Out

$a^{b+c}$  is NEVER  $a^b + a^c$  — you cannot distribute an exponent over addition or subtraction.

Negative exponents create reciprocals, not negative numbers — a common and costly misread.

## Exponent Rules — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

### Exponent Rules — Problem 1 (Very Hard)

Simplify to a single term with positive exponents only:

**Formula / Rule Used:** Power of a product, product rule, negative exponents, and the zero-exponent rule together.

#### Solution

**Step 1:** Recognize that any nonzero expression raised to the power 0 equals 1, so the denominator -- it does not need to be expanded.

**Step 2:** Apply the power rule to the first factor:  $a^{\frac{m}{n} \cdot \frac{p}{q}} = a^{\frac{mp}{nq}}$ .

**Step 3:** Apply the power rule to the second factor:  $b^{\frac{r}{s} \cdot \frac{t}{u}} = b^{\frac{rt}{su}}$ .

**Step 4:** Multiply the two results using the product rule (add exponents of like bases):  $a^{\frac{mp}{nq}} \cdot b^{\frac{rt}{su}}$ .

**Step 5:** Divide by the denominator, which is 1, so the expression is unchanged:  $\frac{a^{\frac{mp}{nq}} \cdot b^{\frac{rt}{su}}}{1}$ .

**Step 6:** Rewrite with positive exponents only:  $\frac{a^{\frac{mp}{nq}} \cdot b^{\frac{rt}{su}}}{1}$ .

**Answer:**  $a^{\frac{mp}{nq}} \cdot b^{\frac{rt}{su}}$

### Tip

When a factor in the expression is raised to the 0 power, simplify it to 1 immediately instead of expanding it -- it never affects the rest of the problem.

### Trap

A student who doesn't recognize  $a^0 = 1$  wastes time expanding it, or worse, mistakenly distributes the 0 as if it were a real exponent to multiply through, instead of just replacing the whole denominator with 1.

### Process of Elimination



| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | Correctly combined all exponents but forgot the last step of rewriting negative exponents as positive -- stopped before flipping to .                            |
|                       | Mis-applied the power rule to , treating as instead of , which lowers the final y exponent from 7 to 5.                                                          |
| -                     | Misinterpreted the negative exponent on the coefficient, computing as -4 instead of the reciprocal , which corrupts the numeric coefficient in the final answer. |

**Shortcut**

Besides recognizing the zero-power factor as 1, there's no shortcut here -- the fastest path is still applying the power rule to each bracket separately before combining, exactly as shown.

**Desmos Tip**

Pick simple values like  $x=2$ ,  $y=1$ , evaluate the original expression and the simplified answer in Desmos's calculator, and confirm they give the same number.

**Exponent Rules — Problem 2 (Hard)**

**Solve for x:**

**Formula / Rule Used:** Rewriting exponential expressions with a common base, then setting exponents equal.

**Solution**

**Step 1:** Rewrite all bases as powers of 3: and .

**Step 2:** Rewrite the left side: .

**Step 3:** Rewrite the right side: .

**Step 4:** Since both sides now have base 3, set the exponents equal:  $4x - 1 = 3x + 3$ .

**Step 5:** Solve for x:  $4x - 3x = 3 + 1$ , so  $x = 4$ .

**Step 6:** Check: left exponent =  $4(4)-1 = 15$ ; right exponent =  $3(4)+3 = 15$ . Both sides equal , confirming  $x = 4$ .

**Answer:**  $x = 4$

**Tip**

Convert every term to the same base first, remembering to fold any leading coefficient (like the 3·) into the exponent before comparing powers.

**Trap**

A student who ignores the leading factor of 3 and sets  $2(2x-1)$  equal directly to  $3(x+1)$  gets  $4x-2=3x+3$ , producing the wrong answer  $x = 5$  instead of the correct  $x = 4$ .

**Process of Elimination**



| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                          |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| $x = 5$               | Ignored the leading factor of 3 outside and set $2(2x-1)$ equal directly to $3(x+1)$ , giving $4x-2=3x+3$ , which solves to $x=5$ .              |
| $x = 2$               | Incorrectly distributed the outer exponent on , writing it as instead of (forgetting to multiply the +1 by 3 as well), then solved $4x-1=3x+1$ . |
| $x = 1.5$             | Misremembered 27 as instead of , giving the right side exponent $2x+2$ instead of $3x+3$ , and solved $4x-1=2x+2$ .                              |

**Shortcut**

There's no faster route than converting to a common base -- but once converted, you can skip fully expanding and just compare the coefficient-adjusted exponents ( $4x-1$  vs  $3x+3$ ) in one line.

**Desmos Tip**

Graph and find their intersection point; the x-coordinate should read exactly 4.

**Exponent Rules — Problem 3 (Hard)**

A positive number  $k$  satisfies . What is the value of ?

**Formula / Rule Used:** Fractional exponents; solve by raising both sides to the reciprocal power.

**Solution**

**Step 1:** To undo the exponent on  $k$ , raise both sides to the reciprocal power, :

**Step 2:** The left side simplifies because :

**Step 3:** Evaluate the right side: . So  $k = 64$ .

**Step 4:** Now compute the requested value, , using  $k = 64$ :

**Step 5:** Evaluate the square root first: .

**Step 6:** Cube the result: .

**Answer:**

**Tip**

When you're given a fractional power of an unknown and asked for a different fractional power of it, solve for the unknown first using the reciprocal-power move, then evaluate the new expression from scratch.

**Trap**

A student may notice that is the reciprocal of and wrongly assume is just , the reciprocal of the given value -- but flipping an exponent does not flip the value;  $k$  must actually be solved for first.

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
|                       | Assumed that since is the reciprocal exponent of , the value must also be the reciprocal of 16, without actually solving for $k$ first. |



|  |                                                                                                                                                                |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Treated the exponent as a multiplier rather than a power throughout, computing as (instead of 64) and then repeating the same error on the final step to get . |
|  | Correctly solved $k=64$ but only took the square root and forgot to then cube the result to finish evaluating .                                                |

**Shortcut**

Since  $k = 64$  works out cleanly, you can track it as means , then directly, without ever writing radical notation.

**Desmos Tip**

Graph and find where  $y = 16$  to read off  $k = 64$ , then evaluate directly in the calculator to confirm 512.

**Exponent Rules — Problem 4 (Very Hard)**

Suppose  $b$  is a positive number such that and . What is the value of ?

**Formula / Rule Used:** Quotient rule for exponents (dividing two equations with a common base), then the zero-exponent rule.

**Solution**

**Step 1:** Divide the second equation by the first to eliminate the unknown exponent structure: .

**Step 2:** Simplify the left side using the quotient rule: .

**Step 3:** So . Solve for  $b$  by taking the fourth root ( $b$  is positive): .

**Step 4:** Substitute  $b = 3$  into the first equation to find  $n$ : , so  $n = 2$ .

**Step 5:** Evaluate the requested expression: .

**Step 6:** Apply the zero-exponent rule: .

**Step 7:** Check by a second route: , confirming the answer.

**Answer:**

**Tip**

When two exponential equations share the same base but different exponents, divide one equation by the other to cancel the base's unknown power and isolate a numeric power of  $b$  you can solve directly.

**Trap**

A student who solves by writing (dividing instead of taking the fourth root) gets a nonsensical base, instead of correctly evaluating .

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                         |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 0                     | Misinterpreted as minus (subtracting values) rather than divided by (subtracting exponents), computing $9 - 9 = 0$ instead of .                 |
| 81                    | Divided the two given equations in the wrong order, getting , then misread the negative exponent as if the equation were , giving instead of 3; |



|   |                                                                                                                                                                   |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | carrying this incorrect base through the rest of the problem produces 81 instead of the correct 1.                                                                |
| 9 | Misread the requested expression as $2^{n-2}$ instead of $2^{n-2}$ , plugging $n=2$ directly into $2^{n-2}$ rather than first evaluating the exponent $n-2 = 0$ . |

**Shortcut**

You can skip solving for  $n$  entirely:  $2^{n-2} = 1$  and  $2^0 = 1$ , so  $n=2$  in two lines.

**Exponent Rules — Problem 5 (Hard)**

**Simplify to a single fraction with positive exponents only:**

**Formula / Rule Used:** Negative exponents and the quotient rule -- distributing a negative outer exponent across a product.

**Solution**

**Step 1:** Apply the outer exponent  $-1$  to every factor inside the parentheses in the denominator:  $\frac{1}{x^2y^4}$ .

**Step 2:** Rewrite the full expression using this result:  $\frac{x^2y^4}{x^2y^4}$ .

**Step 3:** Apply the quotient rule to the  $x$  terms:  $\frac{x^2}{x^2}$ .

**Step 4:** Apply the quotient rule to the  $y$  terms:  $\frac{y^4}{y^4}$ .

**Step 5:** Combine the results:  $\frac{1}{1}$ .

**Step 6:** Rewrite with positive exponents:  $1$ .

**Answer:**  $1$

**Tip**

When an outer exponent applies to a product of two factors, distribute it to both factors' exponents -- including any factor that is already negative -- before simplifying further.

**Trap**

A student may distribute the outer  $-1$  to  $(x^2y^4)$  but leave  $y^4$  unchanged, reasoning it's 'already negative,' getting the wrong intermediate  $\frac{x^2}{y^4}$  instead of the correct  $\frac{1}{x^2y^4}$  -- this flips the final sign on  $y$  and produces the wrong answer  $\frac{x^2}{y^4}$  instead of the correct  $1$ .

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{x^2}{y^4}$     | Distributed the outer $-1$ exponent to $x^2$ but left $y^4$ unchanged (reasoning it was 'already negative'), producing $\frac{x^2}{y^4}$ instead of $\frac{1}{x^2y^4}$ , which flips the sign on $y$ 's exponent in the final step. |
| $\frac{x^2}{y^2}$     | Applied the quotient rule to the $x$ terms as $\frac{x^2}{x^2}$ instead of $\frac{1}{x^2}$ , mishandling the double negative and getting $\frac{x^2}{y^2}$ instead of $\frac{1}{x^2y^4}$ .                                          |
| $\frac{1}{x^2y^2}$    | Subtracted the $y$ exponents in the wrong order ( $4-2$ instead of $2-4$ ) when applying the quotient rule, getting $\frac{1}{x^2y^2}$ instead of $\frac{1}{x^2y^4}$ , which flips $y$ into the                                     |



|  |                        |
|--|------------------------|
|  | numerator incorrectly. |
|--|------------------------|

**Shortcut**

No shortcut beats carefully distributing the outer exponent first -- but once distributed, you can combine both quotient-rule subtractions  $(-3-(-2))$  and  $2-4$  on the same line instead of writing them separately.

**Desmos Tip**

Substitute a value like  $x=2$ ,  $y=3$  into both the original expression and the simplified answer in Desmos's calculator to confirm they evaluate to the same number.

## Exponent Rules — Problem 6 (Very Hard)

A biotech lab in Rotterdam is tracking a bacterial culture that grows according to  $y = a \cdot 2^{bt}$ , where  $t$  is measured in hours and  $a$  is a constant growth factor. The technician recorded that at  $t=2$  hours the population was 20 organisms, and at  $t=5$  hours it was 160 organisms. The lab's petri dish can hold up to 10,000 organisms before needing to be split. Assuming the growth follows this exponential model exactly, what was the population  $N_0$  at  $t=0$ , when the sample was first placed in the dish?

**Formula / Rule Used:** Exponential growth model ; use the quotient rule to eliminate  $N_0$  and solve the resulting equation.

**Solution**

**Step 1:** Translate the two data points into equations:  $20 = a \cdot 2^{2b}$  and  $160 = a \cdot 2^{5b}$ . (The 10,000-organism dish capacity is not needed for this calculation.)

**Step 2:** Divide the second equation by the first to eliminate  $N_0$ :  $\frac{160}{20} = \frac{a \cdot 2^{5b}}{a \cdot 2^{2b}}$ .

**Step 3:** Simplify the left side using the quotient rule:  $8 = 2^{3b}$ .

**Step 4:** So  $2^3 = 2^{3b}$ , which gives  $a = 2$ .

**Step 5:** Substitute  $a = 2$  back into the first equation:  $20 = 2 \cdot 2^{2b}$ , so  $4N_0 = 20$ .

**Step 6:** Solve for  $N_0$ :  $N_0 = 5$ .

**Step 7:** Check:  $20 = 2 \cdot 2^{2(5/2)}$ , which matches the given data.

**Answer:**  $N_0 = 5$  organisms

**Tip**

In a growth word problem with two data points instead of a known starting value, divide the two exponential equations to cancel  $N_0$  first -- that isolates a power of the growth factor  $a$  you can solve directly.

**Trap**

A rushed student divides the equations in the wrong order instead of  $\frac{160}{20}$ , getting  $\frac{20}{160}$ , which leads to  $N_0 = \frac{1}{4}$  and implies the culture is shrinking -- the reverse of what the data actually shows.

**Process of Elimination**

|                       |                         |
|-----------------------|-------------------------|
| Tempting Wrong Answer | Exactly What Went Wrong |
|-----------------------|-------------------------|



|                        |                                                                                                                                                                                                                              |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>80 organisms</b>    | Divided the equations in the wrong order instead of $\cdot$ , getting $\frac{1}{a}$ instead of $a=2$ , then solved $\frac{1}{a}$ instead of 5.                                                                               |
| <b>20 organisms</b>    | Misread the question and reported the population at $t=2$ (which was given directly as 20) instead of solving for $N_0$ , the population at $t=0$ .                                                                          |
| <b>0.625 organisms</b> | After correctly finding $a=2$ , substituted it into the $t=5$ equation but used the $t=2$ population value (20) instead of the $t=5$ value (160), mismatching the exponent and the data point: $\frac{1}{2} \cdot 20 = 10$ . |

**Shortcut**

Dividing the two given equations to eliminate  $N_0$  immediately, as shown, is already the fastest route -- solving for  $N_0$  or  $a$  individually first (e.g., using logarithms) would take longer.

**Desmos Tip**

Graph  $y = N_0 \cdot 2^t$  in Desmos and confirm the curve passes through both (2, 20) and (5, 160) to verify the fitted model before trusting  $N_0 = 5$ .

**Exponent Rules — Problem 7 (Very Hard)****Simplify to a single number:**

**Formula / Rule Used:** Rewriting mixed bases as powers of a common base, then combining fractional and negative exponents.

**Solution**

**Step 1:** Rewrite 27 and 9 as powers of 3:  $27 = 3^3$  and  $9 = 3^2$ .

**Step 2:** Rewrite the first factor:  $3^3 \cdot 3^2 = 3^5$ .

**Step 3:** Rewrite the second factor:  $3^{-1}$ .

**Step 4:** Multiply the numerator's two factors using the product rule:  $3^5 \cdot 3^{-1} = 3^4$ .

**Step 5:** Rewrite the denominator:  $3^2$ .

**Step 6:** Divide the numerator by the denominator:  $3^4 \div 3^2 = 3^2$ .

**Step 7:** Check with a second route: the net exponent of 3 across the whole expression is  $-2 + 3 - (-1) = 2$ , so the value is  $3^2$ , matching.

**Answer:** 9

**Tip**

When a problem mixes different bases with fractional and negative exponents, first rewrite everything as a power of the smallest common base, then combine exponents using the rules -- never evaluate the messy fractional powers by estimation.

**Trap**

A student who tries to evaluate  $3^{-2}$  and  $3^3$  as decimals right away (rather than rewriting both as powers of 3 first) is likely to make an arithmetic slip on the negative fractional exponent and lose track of the final division by  $3^2$ , often forgetting that dividing by  $3^2$  means multiplying by  $\frac{1}{3^2}$ .

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                     | Correctly simplified the numerator to $3^2$ but then divided by $3^2$ as if dividing by 3 (instead of multiplying by 3), computing $3^2 \div 3^2 = 1$ instead of $3^2 \times 3^2 = 9$ . |
| -9                    | Correctly reached $3^{-2}$ but wrongly assumed dividing by a negative-exponent term makes the result negative, writing -9 instead of correctly computing $3^2$ .                        |
|                       | When combining the numerator's exponents, subtracted instead of added them ( $-2-3=-5$ instead of $-2+3=1$ ), giving $3^{-5}$ , then divided by $3^2$ to get $3^{-7}$ instead of 9.     |

**Shortcut**

Since every base reduces to 3, you can skip writing 27 and 9 explicitly and just track the net exponent of 3 in one line:  $-2 + 3 - (-1) = 2$ , so the answer is  $3^2$ .

**Desmos Tip**

Type the original expression directly into Desmos's calculator and confirm it evaluates to exactly 9.

**Exponent Rules — Problem 8 (Hard)**

Suppose  $3^m = 27$  and  $3^n = 9$ , where  $m$  and  $n$  are positive real numbers with  $m + n < 10$  (this bound is not needed for the calculation). What is the value of  $3^{m-n}$ ?

**Formula / Rule Used:** Power and quotient rules applied directly to two given exponential equations sharing a base.

**Solution**

**Step 1:** Rewrite the target expression using exponent rules:  $3^{m-n}$ . (The condition  $m+n<10$  is extra information not needed here.)

**Step 2:** Substitute the known value for the numerator:  $3^2$ , so  $3^{m-n} = \frac{3^2}{3^n}$ .

**Step 3:** Substitute the known value for the denominator:  $3^2$ .

**Step 4:** Divide:  $\frac{3^2}{3^2}$ .

**Step 5:** Simplify the fraction:  $1$ .

**Step 6:** Check: this used only the two given equations combined via the power and quotient rules -- the bound  $m+n<10$  never entered the calculation.

**Answer:** 1

**Tip**

When asked for an expression like  $3^{m-n}$  built from two given exponential equations with the same base, rewrite it as a product or quotient of powers of the two known quantities,  $3^m$  and  $3^n$ , rather than trying to solve for  $m$  and  $n$  individually.

**Trap**



A student may try to solve for  $m$  and  $n$  individually using logarithms ( $m = \log_5(3)$ ,  $n = \log_5(12)$ ) and then compute  $2m-n$  numerically, which is slower and invites rounding errors -- or may get distracted trying to use the irrelevant bound  $m+n < 10$  to set up a system of equations that was never needed.

### Process of Elimination

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                          |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------|
|                       | Misapplied the power rule, treating $5^m$ as $5^{2m}$ instead of $(5^m)^2$ , computing $5^{2m-n}$ instead of $(5^m)^{2-n}$ .     |
| <b>108</b>            | Misread the exponent as $2m+n$ instead of $2m-n$ , multiplying $5^m$ by $5^n$ instead of dividing, getting $9 \times 12 = 108$ . |
|                       | Set up the quotient upside down, computing $5^{m-2n}$ instead of $5^{2m-n}$ .                                                    |

### Shortcut

Since  $2m-n$  breaks cleanly into  $2m$  minus  $n$ , you can substitute directly as  $5^{2m-n} = (5^m)^2 / 5^n$  in one line without ever solving for  $m$  or  $n$  individually -- this is already the fastest method.

## 6. Radicals, Rational Exponents & Scientific Notation

### Radicals, Rational Exponents & Scientific Notation — Ορολογία (Terminology)

Key vocabulary for this skill, in English and Greek — learn these first.

| Term (English)      | Greek — Ελληνικά          |
|---------------------|---------------------------|
| Radical (root)      | Ρίζα (ριζικό)             |
| Rational exponent   | Ρητός εκθέτης             |
| Simplify a radical  | Απλοποίηση ρίζας          |
| Scientific notation | Επιστημονική σημειογραφία |

### Radicals, Rational Exponents & Scientific Notation — Condensed Theory

Radicals and fractional exponents are two notations for the same idea, and scientific notation is exponent rules applied to place value.

#### Radical-Exponent Conversion

##### **$n$ -th root of**

Converting to fractional exponents before combining or simplifying makes mixed radical expressions far easier.

#### Simplifying Radicals

Factor out the largest perfect square (or perfect cube, etc.) hiding inside the radicand.

#### Combining Radicals with Exponents

Convert every term to a fractional exponent with a common denominator, then apply the product/quotient rule, then convert back if needed.

#### Scientific Notation

##### **, where $1 \leq N < 10$**

When multiplying, multiply the coefficients and add the exponents; when dividing, divide the coefficients and subtract the exponents — then renormalize so the coefficient is between 1 and 10.



#### Watch Out

After any scientific-notation multiplication or division, check that your coefficient is still between 1 and 10 — if not, shift the decimal and adjust the exponent by 1.



## Radicals, Rational Exponents & Scientific Notation — 8 Worked Problems

Every problem below is original CLC material written to mirror the twists, traps, and multi-step reasoning of real Digital SAT Hard/Very Hard math questions — not textbook plug-in practice.

### Radicals, Rational Exponents & Scientific Notation — Problem 1 (Hard)

**Simplify completely (assume  $x > 0$  and  $y > 0$ ):** .

**Formula / Rule Used:** Quotient rule for radicals: ; simplify using exponent subtraction.

**Solution**

**Step 1:** Combine the two radicals into one using the quotient rule: .

**Step 2:** Simplify the numerical coefficient inside the radical: .

**Step 3:** Simplify the  $x$  terms using exponent subtraction: .

**Step 4:** Simplify the  $y$  terms using exponent subtraction: .

**Step 5:** The expression under the radical is now: .

**Step 6:** Since 16, , and are all perfect squares, split the radical: .

**Step 7:** Final simplified expression: .

**Answer:**

#### Tip

When you see a quotient of two radicals with the same index, combine them into a single radical first — it is almost always less arithmetic than simplifying each radical separately and then dividing.

#### Trap

Simplifying and separately before dividing invites an exponent bookkeeping error — for instance writing as instead of — which lands on the wrong final answer instead of the correct .

#### Process of Elimination

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                            |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | Subtracted the $y$ exponents incorrectly, computing as instead of .                                                                                                                                |
|                       | Forgot to subtract the denominator's $x$ exponent at all, treating as instead of , while still handling the $y$ terms correctly.                                                                   |
|                       | Correctly simplified inside the radical to and correctly pulled out , but forgot to also take the square root of the numeric coefficient 16, carrying it through unsimplified instead of writing . |

#### Shortcut

Combining under one radical first turns four separate simplifications into a single division, which is already the fastest route on this problem — there is no faster alternate method.

#### Desmos Tip



Plug in a test value like  $x=2$ ,  $y=1$  into both the original expression and the simplified form on Desmos's calculator to confirm they give the same number.

### Radicals, Rational Exponents & Scientific Notation — Problem 2 (Hard)

**Solve for  $x$ :** . State the value(s) of  $x$  that satisfy the original equation.

**Formula / Rule Used:** Isolate the radical, square both sides, solve the resulting equation, and check every solution in the original equation.

**Solution**

**Step 1:** Isolate the radical by adding  $x$  to both sides: .

**Step 2:** Square both sides to eliminate the radical: .

**Step 3:** Move all terms to one side: .

**Step 4:** Factor the quadratic: , so  $(x + 3)(x - 1) = 0$ .

**Step 5:** Solve for  $x$ :  $x = -3$  or  $x = 1$ .

**Step 6:** Check  $x = 1$  in the ORIGINAL equation: . This works.

**Step 7:** Check  $x = -3$  in the ORIGINAL equation: , which does not equal 2. This solution is extraneous.

**Step 8:** Reject  $x = -3$  and keep only the valid solution.

**Answer:**  $x = 1$

**Tip**

After squaring both sides of a radical equation, treat every solution as a candidate only — always substitute back into the ORIGINAL (unsquared) equation before finalizing an answer.

**Trap**

A student who solves the quadratic correctly but forgets to check both roots in the original equation (not the squared one) will report both  $x = 1$  and  $x = -3$ , missing that  $x = -3$  is extraneous because the left side actually evaluates to 4, not 2.

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $x = 1$ and $x = -3$  | Forgot to substitute both candidate roots back into the ORIGINAL (unsquared) equation, so didn't catch that $x = -3$ is extraneous, and reported both roots as valid.                   |
| $x = -3$              | Assumed without checking that the negative-valued root must be the extraneous one, rejected $x = 1$ and kept $x = -3$ instead of actually verifying each root in the original equation. |
| $x = 3$               | Mis-factored as $(x - 3)(x + 1)$ instead of the correct $(x + 3)(x - 1)$ , producing roots $x = 3$ and $x = -1$ rather than the correct $x = 1$ and $x = -3$ .                          |

**Shortcut**



You can eliminate  $x = -3$  without recomputing anything: since  $|x + 2|$  is always non-negative, it can only equal  $x + 2$  when  $x + 2$  is also non-negative ( $x \geq -2$ );  $x = -3$  fails this instantly.

### Desmos Tip

Graph  $y = \sqrt{x + 2}$  and  $y = 2$  and find their intersection point directly — Desmos will show only one true intersection, at  $x = 1$ , immediately revealing that  $x = -3$  is extraneous.

## Radicals, Rational Exponents & Scientific Notation — Problem 3 (Very Hard)

Write  $\sqrt[6]{\frac{x^5}{x^2}}$  as a single power of  $x$  (assume  $x > 0$ ), and simplify to its lowest exponent.

**Formula / Rule Used:** Convert radicals to rational (fractional) exponents with a common denominator, then apply the product and quotient rules for exponents.

### Solution

**Step 1:** Convert  $\sqrt[6]{\frac{x^5}{x^2}}$  to a fractional exponent:  $\left(\frac{x^5}{x^2}\right)^{\frac{1}{6}}$ .

**Step 2:** Convert  $\frac{x^5}{x^2}$  to a fractional exponent:  $x^{\frac{5}{2}}$ .

**Step 3:** Multiply these using the product rule for exponents (add exponents):  $x^{\frac{5}{2} \cdot \frac{1}{6}}$ .

**Step 4:** Find a common denominator of 6 to add the exponents:  $\frac{5}{2}$  and  $\frac{1}{6}$ , so  $\frac{5}{2} = \frac{15}{6}$ .

**Step 5:** The expression so far is  $x^{\frac{15}{6} + \frac{1}{6}}$ .

**Step 6:** Divide using the quotient rule for exponents (subtract exponents):  $x^{\frac{15}{6} - \frac{1}{6}}$ .

**Step 7:** Simplify the exponent:  $x^{\frac{14}{6}}$ .

**Step 8:** Final simplified form:  $x^{\frac{7}{3}}$ .

**Answer:**  $x^{\frac{7}{3}}$

### Tip

Convert every root to a fractional exponent immediately — mixing cube roots and square roots directly is what causes errors, but once everything is written as  $x^{\frac{a}{b}}$ , the problem is just exponent addition and subtraction.

### Trap

A student might try to combine  $\sqrt[6]{\frac{x^5}{x^2}}$  by rewriting both as 6th roots and multiplying radicands directly, which can work but is far more error-prone; the most common slip is treating the division by  $\sqrt[6]{x^2}$  as another multiplication instead of a subtraction, giving  $x^{\frac{15}{6} + \frac{1}{6}}$  instead of the correct  $x^{\frac{14}{6}}$ .

### Process of Elimination

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $x^{\frac{16}{3}}$    | Treated the division by $\sqrt[6]{x^2}$ as multiplication instead of subtraction, computing $\frac{15}{6} + \frac{1}{6}$ instead of $\frac{15}{6} - \frac{1}{6}$ .                                                                            |
| $x^{\frac{14}{3}}$    | Correctly converted and multiplied the two radicals to get $x^{\frac{15}{6}}$ , but forgot to divide by $\sqrt[6]{x^2}$ at all, stopping one step early.                                                                                      |
| $x^{\frac{15}{3}}$    | Added the exponents $\frac{5}{2}$ and $\frac{1}{6}$ by adding numerators and denominators straight across (treated as $\frac{5+1}{2+6}$ ) instead of finding a common denominator of 6, corrupting the exponent before the final subtraction. |



Since all three exponents already share a natural common denominator of 6, you can rewrite the whole expression as a single 6th root from the start: sixth root of = sixth root of .

Graph  $y = \frac{1}{x}$  and  $y = \frac{1}{x^2}$  for  $x > 0$  on Desmos and confirm the two curves lie exactly on top of each other.

A certain nebula has a mass of  $1.5 \times 10^{22}$  kg, spread over a volume of  $1.5 \times 10^{26}$  m<sup>3</sup>. (The nebula was first photographed by an amateur astronomer in 1954.) Find the nebula's average density, in kg/m<sup>3</sup>, in proper scientific notation.

### Solution

**Step 2:** Divide the coefficients: .

**Step 3:** Subtract the exponents: .

**Step 4:** Combine the pieces: density = .

**Step 5:** Since 0.75 is not between 1 and 10, renormalize it: .

**Step 6:** Multiply the powers of 10 together: .

**Step 7:** Final answer: density = kg/.

**Answer:** kg/

Do the coefficient division and the exponent subtraction as two separate steps, then always check whether the resulting coefficient is still between 1 and 10 before calling the answer final.

A student who divides correctly but stops at (since 0.75 is less than 1, it fails the scientific notation rule) reports the wrong final form; a student distracted by the 1954 discovery date might also waste time trying to work it into the calculation.

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| kg/                   | Divided the coefficients and subtracted the exponents correctly, but never renormalized, leaving a coefficient (0.75) that isn't between 1 and 10.                      |
| kg/                   | Subtracted the exponents in the wrong order (47 - 31 instead of 31 - 47), flipping the sign of the exponent and producing a wildly large density instead of a tiny one. |



|     |                                                                                                                                                   |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------|
| kg/ | Made an arithmetic slip computing as 0.8 instead of 0.75 (rounding instead of simplifying), carrying that error through to the final coefficient. |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------|

**Shortcut**

There is no shortcut beyond the standard divide-coefficients/subtract-exponents/renormalize method, but you can skip long division by noticing instantly.

**Desmos Tip**

Type directly into Desmos's calculator and read off the result to confirm it matches after renormalizing.

**Radicals, Rational Exponents & Scientific Notation — Problem 5 (Very Hard)**

If and  $x > 0$ , what is the value of ?

**Formula / Rule Used:** Rational exponent power rule: ; use it either to solve for the base or to jump directly between related exponents.

**Solution**

**Step 1:** Start with the given equation: .

**Step 2:** Raise both sides to the power (the reciprocal of ) to isolate  $x$ : .

**Step 3:** Simplify the left side using the power rule: .

**Step 4:** Simplify the right side: .

**Step 5:** So  $x = 16$ .

**Step 6:** Compute the requested value: .

**Answer:**

**Tip**

When given one rational-exponent value of  $x$  and asked for another, first check whether the target exponent is a simple multiple of the given one before solving for  $x$  from scratch.

**Trap**

A student who treats exponents like linear scaling might compute as times 8, getting , instead of correctly using the power rule ; exponents combine multiplicatively, not by simple proportion.

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                     |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|                       | Treated the exponent relationship as direct proportion, computing , instead of solving for $x$ with the power rule and then re-evaluating . |
| 16                    | Correctly solved for $x$ ( $x = 16$ ) but then stopped, reporting $x$ itself instead of continuing to compute the requested .               |
| 256                   | Confused the exponent with squaring, computing instead of the square root of 16, after correctly solving $x = 16$ .                         |

**Shortcut**

Since , you can skip solving for  $x$  entirely: = — same answer in one line.

**Desmos Tip**

On Desmos's calculator, evaluate directly for a one-step check that matches the value found by fully solving for  $x$  first.

**Radicals, Rational Exponents & Scientific Notation — Problem 6 (Very Hard)**

Aurora Glassworks in Tallinn produces two similar (same-shape) crystal decanters — a display model and a smaller retail model. The display model has a surface area of , and the retail model has a surface area of . If the display model holds of liquid, and volume scales with the cube of the linear scale factor between similar solids, how many of liquid does the retail model hold?

**Formula / Rule Used:** For similar solids, area scales with the square of the linear scale factor and volume scales with the cube; equivalently, volume ratio = .

**Solution**

**Step 1:** Since the decanters are similar, the ratio of their surface areas equals the square of the ratio of their linear scale factors: .

**Step 2:** Simplify the area ratio: .

**Step 3:** Take the square root of both sides to get the linear scale factor: retail linear / display linear = .

**Step 4:** Volume scales with the CUBE of the linear scale factor (not the square used for area), so: retail volume / display volume = .

**Step 5:** Substitute the known display volume: retail volume / .

**Step 6:** Solve for the retail volume: retail volume = .

**Step 7:** Final answer: the retail model holds of liquid.

**Answer:**

**Tip**

Translate word problems about similar solids into scale-factor language first: area ratio = , so take a square root to isolate the linear ratio, then apply the separate power (cube for volume) that the question actually asks about.

**Trap**

A student might apply the same square relationship used for area (ratio ) directly to volume, computing , instead of correctly cubing the linear scale factor to get the true volume ratio of and answer 64.

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | Applied the area ratio directly to volume instead of cubing the linear scale factor, computing instead of .                                             |
|                       | Used the area ratio itself as if it were already the linear scale factor and cubed it directly, skipping the required square-root step that converts an |



|  |                                                                                                                                                               |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | area ratio into a linear ratio: .                                                                                                                             |
|  | Inverted the surface-area ratio (used instead of ), which flipped the scale factor from to 2 and produced a larger volume instead of the correct smaller one. |

**Shortcut**

You can skip finding the linear scale factor as an intermediate decimal and go straight from area ratio to volume ratio using rational exponents: since area ratio is , volume ratio = directly, without ever writing " on its own line.

**Desmos Tip**

Type directly into Desmos's calculator to verify the retail volume equals 64 in a single step.

**Radicals, Rational Exponents & Scientific Notation — Problem 7 (Hard)**

For  $x > 0$ , simplify completely to a single power of  $x$ :  $\sqrt[3]{x^2} \cdot \sqrt{x}$  / .

**Formula / Rule Used:** Convert negative exponents and roots to a single rational exponent, then apply the product and quotient rules by adding/subtracting exponents.

**Solution**

**Step 1:** Convert  $\sqrt[3]{x^2}$  to a rational exponent: .

**Step 2:** Convert  $\sqrt{x}$  to a rational exponent: .

**Step 3:** Combine the numerator using the product rule (add exponents): .

**Step 4:** Find a common denominator of 3: -, so -. The numerator is .

**Step 5:** Divide by the denominator using the quotient rule (subtract exponents): .

**Step 6:** Find a common denominator of 6: - and , so -.

**Step 7:** Final simplified form: .

**Answer:**

**Tip**

Rewrite every negative exponent and every root as a single rational exponent before combining anything — mixing negative-exponent notation with radical notation is what causes sign errors, not the exponent rules themselves.

**Trap**

Subtracting is the same as subtracting a negative, which flips to addition — a rushed student computes - (forgetting to distribute the negative sign onto the denominator's exponent) and lands on instead of correctly writing -.

**Process of Elimination**

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|
|                       | Forgot to distribute the negative sign onto - when subtracting a negative exponent, computing - instead of -, giving instead of -. |



|  |                                                                                                                                                                                            |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | Treated the whole-number exponent -2 as if it already had a denominator of 3 (used - instead of converting it to -) before adding to , corrupting the numerator's exponent from the start. |
|  | Correctly simplified the numerator to but forgot to divide by at all, submitting the numerator's exponent as the final answer.                                                             |

**Shortcut**

Since every term is already a single power of  $x$ , you can add all three exponents at once — treat the denominator's exponent as subtracted with its sign flipped from the start rather than combining the numerator first and the denominator second.

**Desmos Tip**

Plug in  $x = 64$  into both the original expression and on Desmos's calculator (both give 128) to confirm they match numerically.

**Radicals, Rational Exponents & Scientific Notation — Problem 8 (Very Hard)**

A materials lab in Eindhoven measures a thin film's cross-sectional area as times , in square micrometers. Express this area in proper scientific notation.

**Formula / Rule Used:** Power of a product rule: ; apply to each factor, then multiply and renormalize to scientific notation.

**Solution**

**Step 1:** Apply the rational exponent to the first factor: .

**Step 2:** Compute : the cube root of 8 is 2, and 2 squared is 4, so . This factor is .

**Step 3:** Apply the rational exponent to the second factor: .

**Step 4:** Compute , so this factor is .

**Step 5:** Multiply the two simplified factors: .

**Step 6:** Multiply the coefficients:  $4 \times 3 = 12$ . Add the exponents: .

**Step 7:** Combine: . Since 12 is not between 1 and 10, renormalize: .

**Step 8:** Multiply the powers of 10: .

**Step 9:** Final answer: square micrometers.

**Answer:** square micrometers

**Tip**

When a whole scientific-notation quantity is raised to a rational-exponent power, apply that exponent to the coefficient and the power of 10 separately — do not simplify the coefficient and exponent-of-10 in the same step, or you'll mix up which rule applies to which part.

**Trap**



It's tempting to apply the outer exponent (or ) only to the power of 10 and leave the numeric coefficient (8 or 9) unraised, or vice versa — for instance treating as (forgetting to also raise the 8 to the power), which silently skips a required cube-root-then-square step.

### Process of Elimination

| Tempting Wrong Answer | Exactly What Went Wrong                                                                                                                                                    |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | Forgot to raise the coefficient 8 to the power in the first factor, treating as instead of , which threw off the final coefficient.                                        |
|                       | Correctly cube-rooted-then-squared the coefficient 8 to get 4, but forgot to also apply the power to the exponent on 10, leaving instead of correctly scaling it down to . |
|                       | Dropped the negative sign when multiplying the exponent -4 by , writing instead of , which threw the final power of ten off by four orders of magnitude.                   |

### Shortcut

Since both bases already have nice perfect cube/square coefficients (8 and 9) and their powers-of-ten exponents divide evenly by the given fractional exponents, you can compute each factor's coefficient and power of ten in one line each, skipping any written-out radical form entirely.

### Desmos Tip

Type directly into Desmos's calculator and confirm the displayed value matches before writing the final scientific-notation form.



## Quick Reference — Master Summary

Use this page as a final review before test day.

### The 13 Skills at a Glance

| Skill                | First Move                                                            | Main Trap                                                           |
|----------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------|
| Lines                | Find slope first; convert to slope-intercept form.                    | Perpendicular slope: using $m$ instead of $-1/m$ .                  |
| Systems              | Check if you can combine equations directly for the asked expression. | Solving fully for $x$ and $y$ when only a combination was needed.   |
| Inequalities         | Flip the sign only when multiplying/dividing by a negative.           | Forgetting to flip — or flipping when adding/subtracting.           |
| Absolute Value       | Isolate the bars, then split into two cases.                          | Not checking both solutions in the ORIGINAL equation.               |
| Exponents            | Rewrite with a common base before comparing exponents.                | Distributing an exponent over addition — $(a+b)^n \neq a^n + b^n$ . |
| Radicals             | Convert to fractional exponents before combining.                     | Forgetting to renormalize scientific notation coefficients.         |
| Ratios & Proportions | Use the 'x' method — a common multiplier.                             | Answering with the wrong part after solving for $x$ .               |
| Probability          | Identify independent/dependent and mutually exclusive/not first.      | Forgetting the denominator changes without replacement.             |
| Unit Conversions     | Set up the fraction so units cancel diagonally.                       | Forgetting to square/cube the WHOLE conversion factor.              |
| Statistics           | Use $\text{Sum} = \text{Mean} \times n$ to recover totals fast.       | Assuming mean and median move the same way when data changes.       |
| Data Displays        | Read the five-number summary directly off a box plot.                 | Confusing a longer whisker with more data points.                   |
| Scatterplots         | Read predicted values from the LINE, not the data points.             | Using a data point instead of a point actually on the line.         |
| Sampling             | Check random selection first — sample size is secondary.              | Assuming a larger sample fixes a biased selection method.           |

### Golden Rules

- Write down what the question actually asks for LAST — many Hard problems bury a units or sign switch in the final step.
- If a problem gives one extra fact you haven't used yet, check whether you need it before submitting an answer.
- When a problem can be solved two ways, the faster way is usually the one that solves directly for the combined quantity being asked, not each variable separately.



- Use Desmos to confirm an algebraic answer whenever the relationship can be graphed — but never as a substitute for knowing the rule.
- A 'twist' is not a trick question — it's the same rule applied in a less familiar form. Name the rule before you touch the algebra.